# Detecting unknown arbuscular mycorrhizal fungi from linked long-read rRNA gene sequences

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**Abstract**

1. Arbuscular mycorrhizal fungi (AMF) are identified from environmental DNA with three naming systems based on different rRNA gene regions: MaarjAM virtual taxa (VTs; small-subunit, SSU), UNITE species hypotheses (SHs; internal transcribed spacer, ITS) and species clades on large-subunit (LSU) reference trees. The names rarely agree, and rRNA gene copies differ within a genome, so one fungus can appear as several taxa and a divergent copy as a new one. We asked how the systems relate, whether combining them identifies species reliably, and how often species missing from the reference are mistaken for known ones.
2. We named 903 long-read sequences spanning the SSU, ITS and LSU, from 151 cultures of 37 species, in all three systems. We removed each culture and each species from the reference in turn, also removed species from the LSU reference, and tested 52 cultures from other collections and 5,511 EUKARYOME sequences against predictions fixed in advance.
3. SHs were mostly nested within LSU species, and these within VTs: VTs separated at most 19 of 37 species, whereas single cultures spanned up to 6 SHs. In *Rhizoglomus*, sequences with a shortened 5.8S gene shared one deletion across five species; i.e., they are genomic paralogs rather than artifacts. On trees, species formed clades but conspecific cultures did not. With one culture removed, the combined names kept the correct species for 745 of 755 sequences. Species removed from both the culture and LSU references were still identified as known for 7.9% of their sequences (95% interval over species 1.4–17.3%), against 1.3% with the LSU reference intact. These errors fell within species complexes that the markers cannot separate; predictions from the reference flagged every species at risk, but not every misidentified sequence. In cultures from other collections, the only misidentifications (4 of 102 sequences) were those predicted. Of 2,706 EUKARYOME sequences of absent taxa, 13 (0.5%) were identified as known, all of them close relatives of reference species, and 12 of these were not predicted, mostly because they matched reference cultures identified only to genus.
4. Novelty can be inferred from rRNA genes only where the markers separate species, and error rates hold only for the reference databases used. For diversity analyses, divergent copies should be recognized before taxa are counted, and inseparable species should be reported as complexes.

**Keywords.** arbuscular mycorrhizal fungi; Glomeromycota; rRNA gene intragenomic variation; MaarjAM virtual taxa; UNITE species hypotheses; novelty detection; phylogenetic placement; long-read metabarcoding.

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# 1 Introduction

Arbuscular mycorrhizal fungi (AMF; Glomeromycota) live in symbiosis with the roots of about 72% of vascular plant species [4]. Because AMF in roots and soil can rarely be identified by their morphology, their communities are studied mostly by metabarcoding: a marker region of the rRNA genes is amplified from environmental DNA, sequenced, and compared with reference databases [40, 10, 52].

Surveys of AMF name what they find in three incompatible ways, so results from different markers cannot be pooled. There are three well-established methods of identification, all based on the rRNA locus, which holds the genes for the small-subunit (SSU, 18S), 5.8S and large-subunit (LSU, 28S) rRNAs, separated by two internal transcribed spacers, ITS1 and ITS2 (together with the 5.8S gene, the ITS region; Figure 1a). Many SSU-based studies refer to MaarjAM virtual taxa (VTs), i.e., pre-established groups of SSU sequences defined by phylogeny and at least 97% sequence similarity [40, 39]. ITS-based studies commonly use UNITE species hypotheses (SHs), i.e., clusters of ITS sequences at flexible distance thresholds [23, 32]. LSU-based studies usually place reads on curated reference trees and name them after the species clade in which they fall [25, 11, 8, 9]. The ITS and LSU approaches usually give Latin species names, whereas these are mostly lacking for VTs. Apart from the species names shared by the ITS and LSU approaches, there is no conversion from one system to another. Therefore, global occurrence databases such as GlobalAMFungi keep these datasets apart [52]. The codes of the roughly species-level units in the three databases have evolved independently, and many reference sequences are unidentified environmental clones, some of which have been recognized as chimeric [47].

AMF genomes make the comparison harder, because their rRNA gene copies are not homogenized. Copies within one individual differ [41], and a single *Rhizoglomus irregulare* genome carries only ten or eleven complete rRNA gene units, which differ in sequence and are not arranged in tandem [29]. Furthermore, the rRNA gene copies within a culture can differ by several percent in the ITS region [51, 46], and variation in the LSU and SSU genes can also be high (here, within cultures, up to several percent in the LSU but rarely more than 1% in the SSU; Section 3.2 and Methods S3). One genome can therefore look like several taxa, and a divergent copy can look like an unknown lineage or an undescribed species.

Short-read metabarcoding allows only one short marker to be analysed at a time. By contrast, long reads that span the SSU, ITS and LSU link all three markers in a single sequence. Amplicons of about 2.5 kb were introduced to bring the three marker regions into one dataset [22], and a large collection of such sequences from identified cultures now exists [46]. In this collection, every sequence can be named in all three systems at once, and the species name of each culture serves as the known answer for its own sequences. We used it to ask four questions:

1. How do VTs, SHs and LSU-based species clades correspond to each other and to cultured species?

2. To what extent do divergent rRNA gene copies interfere with species recognition and naming?

3. Does combining the three naming systems identify species more reliably than any one of them?

4. When a species is missing from the reference, how often is it misidentified as a known species, and how does this depend on what the reference databases contain?

Throughout, a species is *new* or *unknown* when it is absent from the reference against which a sequence is identified. Our tests remove species from the culture reference and, in one test, from the LSU reference as well. None removes a species from every database, so none measures the

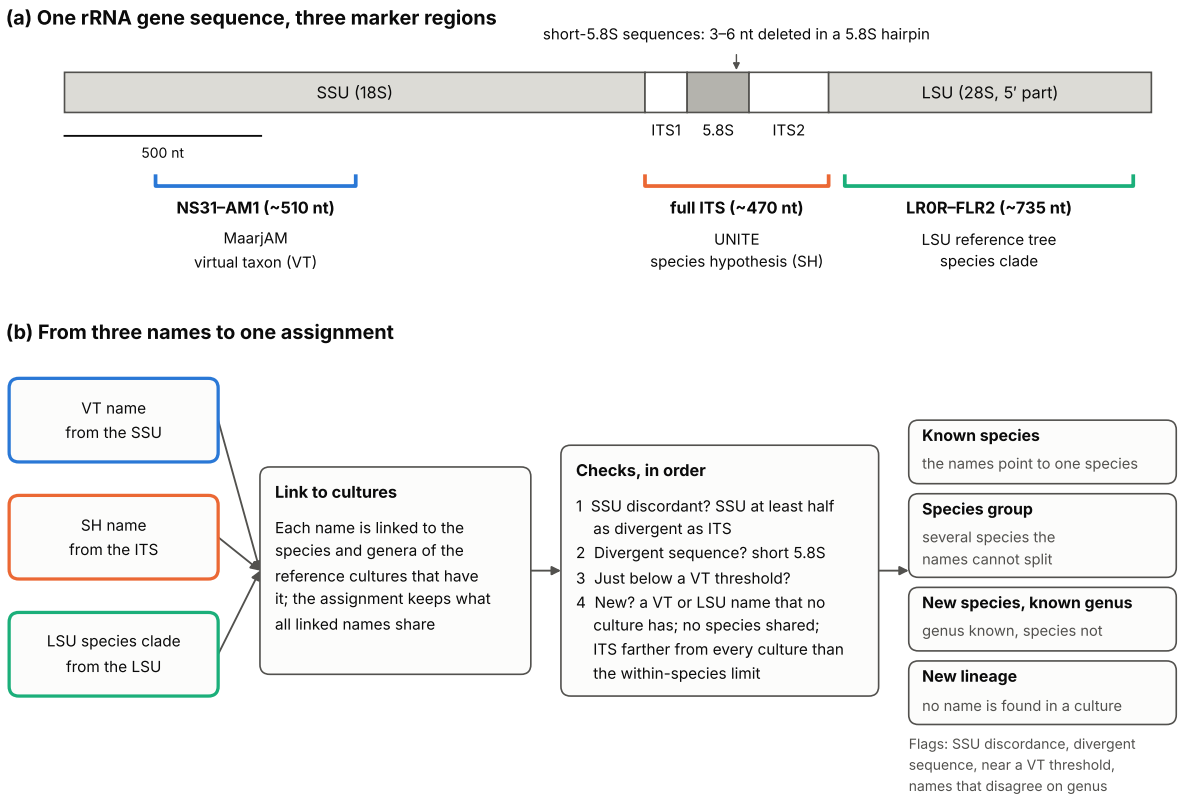



Figure 1: Structure of the rRNA gene sequences and the identification workflow. (a) A long-read sequence drawn at the median coordinates of the 903 culture sequences (2,758 nt), with the three regions on which the naming systems are built: the NS31–AM1 fragment of the SSU for MaarjAM virtual taxa, the full ITS region for UNITE species hypotheses, and the LR0R–FLR2 fragment of the LSU for species clades on the LSU reference tree. The arrow marks the deletion found in the short-5.8S sequences (Section 3.3). (b) Each sequence receives a name in each system, and each name is linked to the reference cultures whose sequences have the same name. The consensus assignment keeps only the taxa that all linked names agree on. Simpler explanations are then tested before a sequence is considered new (Section 2.4).

error for a species that has never been sequenced, and each error rate is reported together with the reference it was measured against. We evaluate the identification of individual sequences, from cultures and from a curated database; we do not evaluate community profiles or richness estimates.

# 2 Materials and methods

## 2.1 Culture sequences and marker regions

We used the 904 rRNA gene sequences from 148 cultures of 44 species deposited by Stefani et al. [46] (GenBank PX207096–PX207274 and PX214440–PX215164). Each sequence is a PacBio read of about 2.8 kb that spans most of the SSU gene, the whole ITS region and the first part of the LSU gene (Figure 1a). We call them sequences rather than copies: each comes from one rRNA gene copy, but two identical sequences may come from the same copy, and a sequence may carry PCR or sequencing errors. Stefani et al. also sequenced three protein-coding genes, which we used only to examine species that the rRNA genes cannot separate (Section 4.6). Three culture labels hold two lineages each (mixed cultures), which we treated as separate cultures,

yielding 151 cultures in total. We used the taxon names as deposited, except that species deposited in *Rhizophagus* are given here in *Rhizoglomus* [44], the name we use for this genus (e.g., *Rhizoglomus irregulare* for *Rhizophagus irregularis*); the databases use either name, and we treated the two as synonyms throughout.

In each sequence, ITSxRust [33] located the SSU, ITS1, 5.8S, ITS2 and LSU, and we extracted, at the original primer sites, the fragment on which each database was built: NS31–AM1 for MaarjAM [45, 17, 40], the full ITS region (ITS1, 5.8S and ITS2) for UNITE, and LR0R–FLR2 for the LSU reference [5, 49, 11]. Four additional marker fragments used in AMF studies were extracted in the same way (Table S1; Methods S1). For phylogenetic trees and rRNA structure, we used the whole length of each sequence, split into three regions: the *SSU region* (all SSU sequence in the read), the full ITS region, and the *LSU region* (all LSU sequence in the read).

The culture reference, i.e., the curated set of culture sequences against which other sequences are compared, was built from the deposited sequences and a list of curation edits (Table S2; Methods S2). Each edit was written together with a check of what the data had to look like, and the build stopped if any check failed. One sequence was removed as a probable error or pseudogene (Section 3.1). The final reference (version 4) holds 903 sequences of 151 cultures and 41 species: 37 named species and 4 species identified to genus only.

Within a culture, most sequences share one main rRNA gene type. We sorted the sequences of each culture into three classes by their 5.8S gene, comparing each with the culture's medoid, i.e., the sequence closest to all others of its culture:

1. *main-type* sequences, whose 5.8S differs by at most 3% from that of the medoid;
2. *short-5.8S* sequences, whose 5.8S differs by more than 3% and is at least 3 bp shorter than that of the culture's main type;
3. *same-length* sequences, whose 5.8S differs by more than 3% but is not shorter.

Short-5.8S and same-length sequences together are called *divergent* sequences.

The distance between two sequences was the number of substitutions, insertions and deletions needed to align the shorter sequence within the longer one (the edit, or Levenshtein, distance, with free end gaps in the longer sequence), expressed as a percentage of the length of the shorter sequence (computed with edlib [54]).

## 2.2 Naming each sequence in the three systems

Each sequence was searched with BLAST+ [6] against the database of each system, with marker-specific identity and coverage thresholds (Table 1): its SSU region against MaarjAM (the gDAT snapshot of July 2020 [40]), whose references span the NS31–AM1 fragment; its full ITS region against UNITE 10.0 (the full UNITE+INSD set of 19 February 2025 [32, 1]); and its LR0R–FLR2 fragment against the LSU reference of Delavaux et al. (V18 of May 2025 [11, 8, 9]). Hits to any sequence of the culture reference were disregarded, so that no sequence could take its name from itself or from its own culture.

The best BLAST hit can change with one or two differences in a sequence. Each sequence therefore kept not only the name of its best hit but every name whose hit was almost as good, within a margin of identity of the best hit; we call these its *near-best names*. For each system, the margin was the smallest at which the main-type sequences of at least 99% of cultures share at least one name. In addition, where the main-type sequences of one culture were split between two names, and the sequences on either side of the split differed by less than 1% in the SSU, the two names were merged (Table S3; Methods S3).

For the LSU, sequences were also placed on the V18 reference tree with EPA-ng [2]. A placement within a clade that holds records of a single V18 species was read as that species (the *LSU species clade*; Methods S4).

Table 1: BLAST+ searches with which each of the 903 culture sequences was named in the three systems (e-value at most $10^{-50}$ in every search).

| System | Region | BLAST task | Database | Identity | Coverage |
|---|---|---|---|---|---|
| VT | SSU region | megablast | MaarjAM (gDAT, July 2020; 3,081 seq., 384 VT) | $\geq 97\%$ | $\geq 95\%$ |
| SH | full ITS | blastn | UNITE 10.0, UNITE+INSD (16,800 Glomeromycota seq., 7,563 SH) | $\geq 98.5\%$ | $\geq 90\%$ |
| LSU | LR0R–FLR2 | blastn | Delavaux V18 (460 AMF records, 234 species labels) | $\geq 97\%$ | $\geq 90\%$ |

One region of each sequence is searched against one database per system. Identity and coverage are the minimum values for a hit to pass. Coverage is the share of the shorter of the two sequences that the alignment covers, not counting Ns in the reference. MaarjAM reference sequences span only the NS31–AM1 fragment. Hits to the culture sequences themselves were ignored. The SH search was also run against the UNITE general release, which keeps one representative sequence per SH (2,048 for Glomeromycota; Section 3.2).

### 2.3 Combining the three names

Each name in each system (a VT, an SH or an LSU species clade) was linked to the species and genera of the reference cultures whose sequences have the same name (Figure 1b). A name was *matched* when at least one reference culture has it and *unmatched* when none does; a sequence with no BLAST hit passing the thresholds of a system had no name in that system. The *consensus assignment* kept only what the matched names agree on: the genera common to all matched names, and, among the matched names linked to named species, the species common to them. For example, a VT shared by cultures of *Glomus chinense* and *Glomus rugosae*, together with an LSU species clade held only by *Glomus rugosae*, gives *Glomus rugosae*. The outcome was a species; a species group (several species that the names cannot separate); a genus only, when the matched names agree on a genus but not on a species; *discordant*, when the matched names point to different genera; or no assignment, when no name was matched.

### 2.4 Deciding when a sequence is new

A sequence was considered new when its species, or its lineage, is not represented in the culture reference. Before a sequence was considered new, three simpler explanations were tested in a fixed order (Figure 1b; Methods S5):

1. **SSU discordance (possible artifact).** Compared with the nearest main-type sequence of another culture, the SSU differs by at least 1% and by at least half as much as the full ITS. Between cultures the SSU changes far more slowly than the ITS (Section 3.4), whereas errors spread along a read, and copies freed from selection, change both at similar rates. The flag marks an SSU unlike that of any cultured relative; it is not a chimera screen (Section 3.4).

2. **Divergent sequence.** The sequence's 5.8S is at least 3 bp shorter than, and at least 3% different from, the main-type 5.8S of the nearest culture; or its best VT or SH is reached, in at least two other cultures, only by short-5.8S sequences (a *paralog name*). Such sequences are named by their VT alone, because their SSU is that of their genome. A same-length 5.8S that differs by 3% or more is only flagged.

3. **Near a VT threshold.** The sequence has no VT name, but its best VT hit lies within one percentage point of the 97% threshold.

4. **New.** An assignment to a species was changed to *new species in a known genus* when the VT or the LSU name is unmatched, when the matched names share a genus but no

species, or when the full ITS lies farther from every reference culture than the within-species limit of the ITS (Section 2.5; the *ITS distance rule*). A sequence none of whose names is matched was classified as a *new lineage*, which has no genus. An unmatched SH does not change an assignment, because the rRNA gene copies of a single genome often fall into several SHs (Section 3.2).

### 2.5 Species complexes

Some named species cannot be distinguished by any of the three markers, and we identified these from the reference alone. For each marker (full ITS, LR0R–FLR2 and SSU region), we measured the distance from each main-type sequence to the nearest main-type sequence of another culture of the same species, and took the 99th percentile of these distances as the *within-species limit*: 6.0% in the ITS, 3.0% in the LSU and 0.8% in the SSU. Two named species were considered *inseparable* when, on every marker, their closest sequences were no farther apart than this limit, so that a sequence of one falls within the variation of the other. Inseparable pairs that share a species were joined into a *species complex*, which is reported as one unit, a species group, wherever a species is assigned. The rule uses no test result and adds no threshold beyond the percentile already used by the ITS distance rule. Because that percentile was chosen after the distances in these cultures had been seen, the ITS distance rule and the complexes were also recomputed together at the 95th and 97.5th percentiles (Methods S6).

The rule can be used in two ways. Applied to the species of the reference, it defines the complexes, a retrospective statement about these cultures. Applied to a new sequence, it says whether that sequence lies *inside* a reference species, i.e., within the within-species limit of that species on all three markers. If the complexes mark the limit of what the markers can resolve, only such a sequence can be misidentified as that species. This use needs no test result, and we used it to fix the predictions of the two external tests before they were run (Sections 2.7 and 2.8). After review, we also applied it within each round of the LSU-removal test (Section 2.6), with the withheld cultures removed and the limits recomputed without them (Methods S6).

### 2.6 Tests with withheld cultures and species

We evaluated the assignments with leave-one-out tests: part of the reference was removed (withheld), and the withheld sequences were identified against the rest. In each test round, everything learned from the cultures (the merged names, the margins, the links from names to species, the comparisons of the divergent-sequence test, the paralog names and the limit of the ITS distance rule) was recomputed from the remaining cultures only (Methods S6).

*Withheld cultures.* In 148 rounds, one culture label was withheld, together with both lineages where a label holds a mixed culture. We asked whether the main-type sequences of species still represented by another culture were assigned to a known species or species group that includes their own species.

*Withheld species.* In 50 rounds, one named species was withheld with all its cultures (37 rounds), or one culture not named to species (13 rounds). A *false-known assignment* is an assignment to a known species or species group for a sequence whose species has no culture left in the reference.

*Withheld from the LSU reference as well.* In the tests above, the three databases stayed intact, so a withheld species could still have its own VT, SH or LSU species clade. In the *LSU-removal test*, each withheld species was also removed from the LSU reference: the reference alignment was rebuilt without its records, the model refitted and all sequences placed again. MaarjAM and UNITE were not edited, so a withheld species kept any VT or SH it has there; this matters because an unmatched VT changes an assignment to 'new' and an unmatched SH does not. This test is the closest of ours to a species that has never been sequenced, but it does not equal one. The ITS distance rule was added after two species, *Glomus chinense* and

*Glomus rugosae*, had been run in this test; we report these as *development species*, because they were used to develop the rule. The 34 other species whose LSU records could be removed (the *test species*) were run after the rule had been fixed; *Ambispora callosa*, which the LSU reference does not hold by name, is reported apart. In 15 further rounds, all cultures of a genus and the genus's LSU records were withheld together.

*Uncertainty.* Sequences of one culture, and cultures of one species, are not independent test cases. The main error rates are therefore given with 95% intervals from cluster bootstraps that resample species, or cultures, with replacement (10,000 replicates), together with the numbers of species and cultures that contribute errors (Table 12; Methods S6).

These cultures were also used to develop the methods, so every result from them is a development estimate.

### 2.7 Cultures from other collections

To test the method on cultures not used to develop it, we used LongGloDB v1 (July 2026) [14], which holds curated SSU–ITS–LSU sequences of AMF cultures from several collections, given as amplicon sequence variants (ASVs), i.e., exact sequence variants inferred from long-read amplicons. After removing the ASVs of the Stefani et al. cultures and of cultures sharing their culture codes (Methods S7), the test set holds 280 ASVs of 52 cultures from three other laboratories. Of these, 164 ASVs (22 cultures) belong to species present in our reference; 59 belong to species absent from the reference but in genera present in it; 43 belong to genera absent from the reference; and 14 are identified to genus only. The absent species are represented by 26 cultures of 19 species.

Before any test ASV was searched, placed or scored, we wrote down the predictions of the complex rule (Section 2.5) for every test ASV, together with a protocol and two amendments that set the outcomes and analyses. We fixed these files by publishing their checksums, i.e., digital fingerprints that reveal any later change (Methods S7); this amounts to a pre-registration. Under the rule, a test ASV can be wrongly identified as a known species only if it lies inside that species on all three markers, so a false-known assignment outside the predicted set would contradict the result of Section 3.7. The rules were used unchanged and learned only from the 151 reference cultures, so no test ASV influenced them. The test ASVs were processed like the culture sequences and named against the same database releases, first with the LSU reference intact and then with each of the 19 absent species removed from it in its own round, as in the LSU-removal test.

The test ASVs begin about 60 bp inside the NS31–AM1 fragment, so they rarely cover the 95% of a MaarjAM reference that the VT search requires: only 33 of the 280 ASVs had a VT hit that passed. We therefore ran two analyses, both fixed in advance: the main analysis applied the rules as specified, and a second analysis counted VT coverage only over the part of each MaarjAM reference that the ASV spans, changing nothing else.

### 2.8 Sequences from EUKARYOME

The species absent from the LongGloDB test were mostly distant from those of the reference. To test close relatives of reference species, and absent lineages in large numbers, we used the long-read sequences of EUKARYOME v2.1 [47]. These are environmental and reference sequences that span the SSU, ITS and LSU and carry EUKARYOME's classification rather than a culture name. Of its 5,546 Glomeromycota sequences whose three regions each span at least 300 nt, 5,540 were processed like the culture sequences. Their names were matched to those of the reference with a fixed table, written before any sequence was compared with anything: names in *Rhizophagus* were read as *Rhizoglomus*, epithets were matched on their stem, and seven names whose relation to the reference is uncertain were decided one by one, each with its reason recorded. EUKARYOME's placeholder lineages, which have no named member, count as genera

that the reference does not hold. We removed the 19 sequences whose full ITS is identical to a sequence of the reference (4 of them rRNA gene copies from genome assemblies), and 10 others that lacked a genus or one of the three regions. The test set holds 5,511 sequences: 199 of 20 species present in the reference; 77 of 33 species absent from it, in genera present in it; 2,629 of lineages absent from it (2,099 of 27 named genera and 530 of 36 placeholder lineages); and 2,606 identified only to a genus present in the reference, which cannot be scored. None is identical in full ITS to a LongGloDB ASV. EUKARYOME's names are a classification, not culture names, and are less reliable at species level; they are independent of the method, but EUKARYOME is curated by one of us (L.T.).

As in the LongGloDB test, the predictions of the complex rule and a protocol fixing the outcomes, conditions and analyses were fixed before any sequence was searched, placed or scored (Methods S7). With the within-species limits of Section 2.5, only 1 of the 2,706 sequences of absent taxa lies inside a reference species (*Diversispora bareai* inside *Diversispora arenaria* and *Diversispora jakucsiae*). Each absent species that V18 holds was removed from the LSU reference in its own round (30 rounds), and each absent genus that V18 holds with all its records (21 rounds); the 583 sequences of taxa that V18 does not name, 530 of them in placeholder lineages, were scored with the LSU reference intact. Most sequences start 21 nt inside the NS31–AM1 fragment, close to the coverage that the VT search requires, so the VT search was run as specified and, in a second analysis also fixed in advance, with coverage counted over the part of each MaarjAM reference that a sequence spans. Any genus given to a sequence of an absent genus counts as wrong, except for *Nanoglomus*: 42 of its sequences lie inside reference cultures named only as *Glomus* sp. on all three markers, so these cultures may be *Nanoglomus*. After the results were in, we examined the false-known assignments outside the prediction by measuring their distance on each marker to every reference culture, including those identified only to genus (`analysis/euk_followup.py`); this analysis is exploratory.

### 2.9 Phylogenetic trees, artifact check and rRNA structure

The phylogenetic trees, the artifact check and the structure analysis were specified in a protocol whose checksum was published before they were run; the structure analysis was added in a dated addendum, likewise fixed before its results were seen (Methods S8).

*Phylogenetic trees.* For each taxonomic order, identical sequences were collapsed and the sequences aligned with MAFFT L-INS-i [21], separately for the SSU region, the full ITS and the LSU region, and for the three joined end to end (the *concatenated sequence*). Maximum-likelihood trees were inferred with IQ-TREE 2 [30] with ModelFinder [20], 1,000 ultrafast bootstraps [18] and 1,000 SH-aLRT replicates [16]. A culture, named species or complex was *monophyletic* when one branch of the unrooted tree separated exactly its sequences from all others, and *strongly non-monophyletic* when a branch that contradicts it had SH-aLRT $\geq 80$ and ultrafast bootstrap $\geq 95$. Each group was tested with all sequences and with the divergent sequences removed.

*Artifact check.* Each divergent sequence was aligned to the most similar main-type sequence of its own culture. A deletion was *long* when it reached, in its region, the 99th percentile of the longest deletion between main-type sequences of one culture, with a floor of 3 nt: 3 nt in the SSU, 5.8S and LSU, and 5 nt in ITS1 and ITS2. A divergent sequence *recurred* when a sequence within 1.0% of it in full ITS occurred under another culture label, i.e., in another isolate. The verdicts were fixed in advance: a recurring sequence without long deletions is a paralog; a recurring sequence with long deletions is a recurrent deleted copy (genomic, possibly a pseudogene); a single sequence with long deletions is a probable artifact; and a single sequence without them is unresolved.

*rRNA structure.* To ask whether divergent sequences keep the structure of functional rRNA, the SSU, 5.8S and LSU of every sequence were aligned with Infernal [31] to the Rfam models of eukaryotic SSU rRNA (RF01960), 5.8S rRNA (RF00002) and eukaryotic LSU rRNA (RF02543)

[38], which describe both the sequence and the secondary structure of each RNA family. A base pair of the consensus structure was *broken* in a sequence when the two bases could not pair (not a Watson–Crick or G·U pair) or when one of them was deleted. A sequence was *degraded* in a region when its share of broken base pairs exceeded the 99th percentile among main-type sequences, and *irregular* when its substitutions, or its insertion and deletion events, counted against the most similar main-type sequence of its culture, exceeded the 99th percentile among main-type sequences. The verdict for the short-5.8S class was fixed in advance: *structurally intact* if at most 10% of its sequences were degraded in each region and its shared deletion removed only unpaired positions of the consensus structure; *structurally degraded*, as expected of pseudogenes, if at least 50% were degraded in some region or the shared deletion removed paired positions; and *mixed* otherwise. To compare the folding of the ITS2 between classes, the ITS2 of every sequence was folded with RNAfold [28]; this comparison was descriptive and entered no verdict.

*Controls for the SSU-discordance flag.* After review, we asked what the first decision step detects, using sequences whose origin is known: real main-type sequences, sequences given random errors, and chimeras built from sequences of two species (Methods S8). This analysis is exploratory and changes no rule.

### 2.10 Distances at each level of relatedness

To explore similarity thresholds without fixing any rule, we measured, for every main-type sequence, the distance to its nearest main-type sequence in four classes: the same culture, another culture of the same species, another species of the same genus, and another genus of the same order. We did this for the NS31–AM1 fragment, on which VTs are defined, the whole SSU region, the full ITS and LR0R–FLR2, and we also measured the distances between the medoid sequences of every pair of cultures within an order. The same distances were measured on two longer regions, the Krüger fragment and the whole sequence. Finally, we counted what the short-5.8S type does to clusters and to the tree of each marker, and tested a sequence motif that marks it (Methods S8).

## 3 Results

### 3.1 The culture sequences

ITSxRust located the regions in all 904 deposited sequences, and every fragment whose primers lie inside a sequence was recovered (Table S1). The fragments are close in length to the amplicons on which the databases were built: NS31–AM1 has a median length of 508 bp without its primers and about 550 bp with them, the size of the fragment behind MaarjAM virtual taxa. The 28 sequences without an NS31–AM1 fragment are exactly the Paraglomerales sequences, in which the AM1 primer does not bind.

Curation changed little (Table S2). One sequence was removed because its SSU and ITS differ from the other sequences of its culture at the same rate, the signature of an error or a pseudogene. One culture held a second *Paraglomus* lineage, and two *Septoglomus* cultures, SO43BL and Blasz_17_2277A, carry the same two lineages in reversed proportions; these were split. The two *Septoglomus* lineages differ as species do (ITS1 11–12%, 5.8S 5.7%, LSU 1.6%). Two *Rhizoglomus irregulare* cultures were kept under their name although their SSU falls in VTX00113, a VT that also holds *Rhizoglomus vesiculiferum*, because their ITS and LSU are closer to *Rhizoglomus irregulare* (in T3_IVT34A, the ITS2 lies a median 3.3% from the nearest *Rhizoglomus irregulare* sequence and 6.4% from the nearest *Rhizoglomus vesiculiferum* sequence). In two cultures, the medoid fell among the short-5.8S sequences, which inverted the classes; we set them by 5.8S length. In one of these, SD5_IVT32, the corrected main type matches the V18 record of *Rhizoglomus proliferum* at 98.5–99.9% identity, whereas its short-5.8S sequences are

91.8% from anything in V18. The final reference holds 903 sequences of 151 cultures of 37 named species, with 51 divergent sequences: 43 short-5.8S sequences in 24 cultures and 8 same-length sequences in 5 cultures.

### 3.2 The three naming systems are nested

The SSU gave a VT name to 895 of the 903 sequences. Counting each culture by its most common VT, 3 of 37 named species span more than one VT, and 8 of 30 VTs hold more than one species; VTX00199 alone holds four *Glomus* species. The nearest LSU reference record gives a culture its own species for 105 of the 135 named cultures whose species V18 holds (78%; 24 get a congener and 6 no name), and the correct genus for all of those that get a name. V18 itself, however, holds two records for nearly every species (460 records under 234 species names: 215 names with two records, five with three or four, and 14 with one). Of the 220 names with two or more records, 45 hold two records that are less than 97% identical, and when each record was left out in turn, only 72% of records returned their own species.

UNITE+INSD, which holds all public sequences, named 592 sequences (66%), against 383 (42%) for the general release, which keeps one representative per SH. An SH is narrower than a genome: the main-type sequences of 74 of the 107 cultures with two or more SH names fall in 2–6 SHs. Counting each culture by its most common SH, 0 of 70 SHs hold two named species, and VTs split 0 of 77 SHs. Once the SHs of one genome are merged into one group, one group is shared by *Glomus macrocarpum* and *Glomus rugosae*. The LSU splits 1 of 73 SHs (into the V18 species *Diversispora varaderana* and *Diversispora bareai*), and VTs split 4 of 40 LSU names. Thus, SHs are mostly nested within LSU species, which are mostly nested within VTs (Figure 2).

Within a culture, the SSU hardly varies (median 0.07%, 95th percentile 0.54%, 99th percentile 0.95%). Merging two pairs of VTs, VTX00113 with VTX00115 and VTX00125 with VTX00222, gives every culture one consistent set of VT names (Table S3), and the margin needed to absorb the SSU differences within cultures (0.6 percentage points) costs three species of resolution. SHs merged into one group per genome separate 32 of the 37 species, against 16 for VTs; this is an upper bound, because the merges were learned from the same cultures.

### 3.3 Divergent sequences, and a paralog family in *Rhizoglomus*

The two divergent classes mislead each marker differently (Table 2). Short-5.8S sequences are well represented in public data, as expected for the variable, non-tandem rRNA gene units of a single genome [29]: their median best SH hit (99.8%) is closer than that of main-type sequences (99.1%). Divergent sequences reach 14 SHs, of which 11 are reached by no main-type sequence (paralog SHs, 4 of them under placeholder names) and 3 by one main-type sequence each (Table 3). The short-5.8S sequences of *Septoglomus viscosum* are the only culture sequences that reach VTX00409, about two percentage points closer than to VTX00063. In *Gigaspora rosea*, four cultures carry the same three or four rRNA gene types, identical across cultures and up to 6.2% apart in the LSU, each in its own SH named *Gigaspora rosea*. These types are not confined to one isolate held in several collections: in LongGloDB (Section 3.8), INVAM cultures under other accession codes (BR155B, FL105 and KS885; FL105 is INVAM's reference culture for the species) carry ASVs within 0.25% of them in full ITS.

Under the artifact rule fixed in advance (Table 4), 3 same-length sequences are paralogs (the *Gigaspora rosea* types, which match main-type sequences of other cultures), 22 short-5.8S sequences are recurrent deleted copies, and 21 short-5.8S and 3 same-length sequences are probable artifacts. The rule counts a type as recurring only when its whole ITS recurs, and many short-5.8S sequences fail that test because their spacers differ. Their deletion, however, recurs. In a reading made after the check, every one of the 43 short-5.8S sequences, including the 21 probable artifacts, carries a 5.8S deletion of the same length, starting within 2 nt, as a sequence under another culture label. In *Rhizoglomus*, all 40 short-5.8S sequences share one 6-nt deletion,

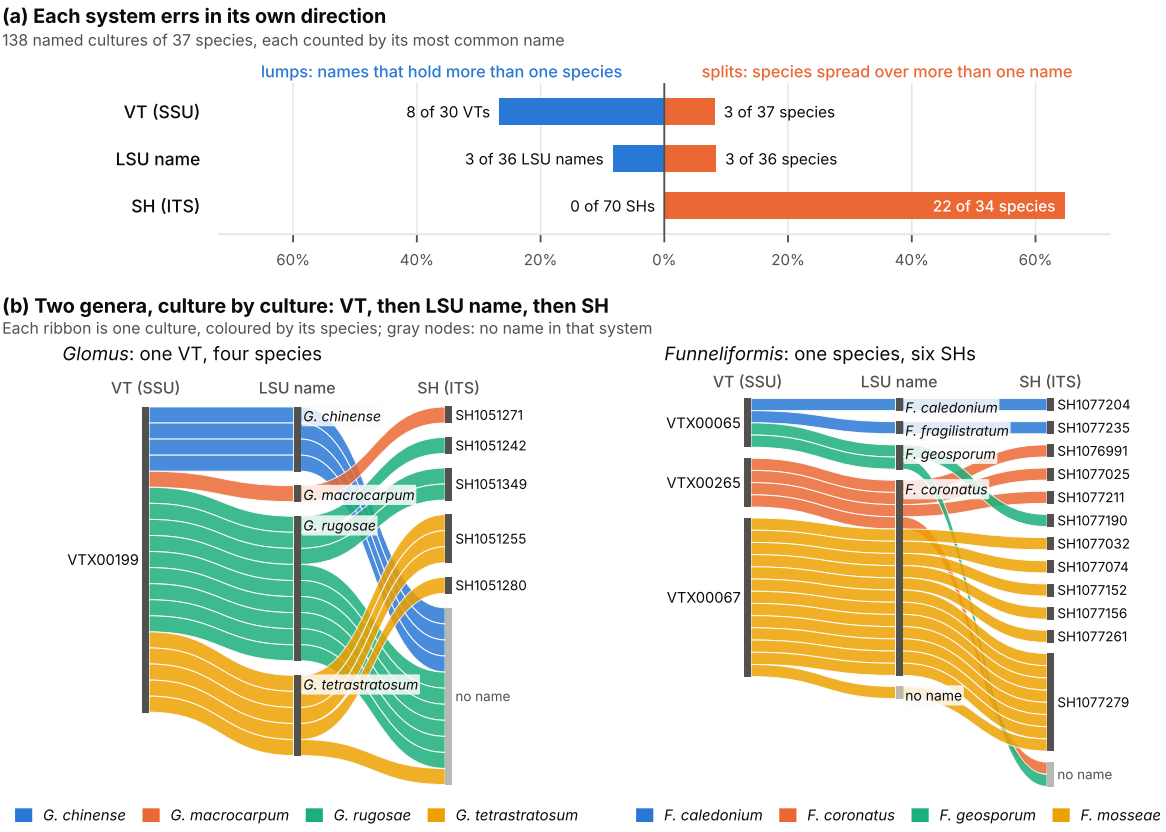



Figure 2: Correspondence between the three naming systems and cultured species. (a) Over the 138 named cultures of 37 species, each counted by its most common name in each system: the share of names that hold more than one species (left) and the share of species whose cultures fall under more than one name (right). VTs lump species and SHs split them; LSU names rarely do either. (b) The cultures of two genera, followed from their VT through their LSU name (the nearest V18 record) to their SH; each ribbon is one culture, coloured by its species. All four *Glomus* species share VTX00199, and 11 of the 19 *Glomus* cultures get no SH. Twelve of the 13 *Funneliformis mosseae* cultures get the LSU name of *Funneliformis coronatus*, and they fall into six SHs; *Funneliformis caledonium* and *Funneliformis geosporum* share VTX00065.

Table 2: Names given to the two classes of divergent sequences in the three systems, compared with the main type of their culture.

| | Short 5.8S | Same length |
|---|---|---|
| Sequences (cultures) | 43 (24) | 8 (5) |
| VT: same as the culture's main type / other / no name | 35 / 7 / 1 | 8 / 0 / 0 |
| LSU: same species / another species / no name | 5 / 1 / 31 | 2 / 5 / 1 |
| LSU best-hit identity, median (main type 98.9%) | 94.9% | 97.2% |
| SH: same / other SH / no name | 0 / 28 / 7 | 0 / 4 / 1 |
| SH best-hit identity, median (main type 99.1%) | 99.8% | 98.6% |

Each divergent sequence compared with the most common name of its culture's main-type sequences (culture reference version 4). Sequences in cultures without a main-type name in a system are left out of that system's row (short 5.8S: VT 0, LSU 6, SH 8; same length: VT 0, LSU 0, SH 3).

Table 3: SHs and a VT that are reached by divergent sequences but by at most one main-type sequence (paralog names).

| Taxon code | Taxon of the best hit | Divergent sequences | Main[a] | Identity[b] |
|---|---|---|---|---|
| SH1337353.10FU | *Rhizoglomus irregulare* | *Rhizoglomus irregulare* 10 (7 cultures) | 1 | 100.0% |
| SH1364117.10FU | Glomeromycota sp. | *Rhizoglomus vesiculiferum* 7 (2 cultures) | 0 | 98.5–99.8% |
| SH1081549.10FU | *Rhizoglomus* sp. | *Rhizoglomus proliferum* 4 (1 culture) | 0 | 99.3–99.8% |
| SH1081557.10FU | *Rhizoglomus intraradices* | *Rhizoglomus intraradices* 4 (3 cultures) | 0 | 100.0% |
| SH1076762.10FU | *Gigaspora margarita* | *Gigaspora rosea* 2 (1 culture) | 1 | 99.8% |
| SH1337340.10FU | *Rhizoglomus* sp. | *Rhizoglomus* sp. 1; *Rhizoglomus irregulare* 1 (2 cultures) | 0 | 100.0% |
| SH1337458.10FU | *Rhizoglomus irregulare* | *Rhizoglomus silesianum* 2 (1 culture) | 0 | 100.0% |
| SH1389224.10FU | Glomeraceae sp. | *Septoglomus viscosum* 2 (1 culture) | 0 | 100.0% |
| VTX00409 | environmental (MaarjAM) | *Septoglomus viscosum* 3 (2 cultures) | 0 | 98.5–99.4% |

[a] Main-type sequences of any culture with that name. [b] BLAST identity of the divergent sequences to the best hit. Taxa are those of the best UNITE record (MaarjAM's for the VT). 6 further names take one divergent sequence each (SH1076712, SH1196937, SH1337247, SH1337463, SH1337480, SH1364178).

found in 23 cultures of five named species and *Rhizoglomus* sp.; so does one sequence classified as main type (PX214776.1, *Rhizoglomus irregulare* 4569_1185B, the main-type sequence in SH1337353; Table 3). The three same-length probable artifacts rest on ITS2 deletions of 8–9 nt; if only deletions of 10 nt or more counted as long (a sensitivity analysis fixed in advance), none would be a probable artifact. A deletion shared across species and isolates is not a PCR or sequencing error, so the short-5.8S sequences of *Rhizoglomus* represent a genomic paralog family rather than artifacts.

Table 4: Divergent sequences by whether the same sequence type occurs in other cultures (recurring) and whether it has long deletions.

| | Recurring | | Single | |
|---|---|---|---|---|
| Class (sequences) | intact | deleted | intact | deleted |
| Short 5.8S (43) | 0 | 22 | 0 | 21 |
| Same length (8) | 3 | 0 | 2 | 3 |

*Recurring*: a sequence within 1.0% in full ITS occurs under another culture label. *Deleted*: compared with the most similar main-type sequence of the same culture, the sequence lacks at least 3 nt in a row in the SSU, 5.8S or LSU, or 5 nt in ITS1 or ITS2 (longer than 99% of such gaps between main-type sequences). The four cells correspond to the verdicts fixed before the check: paralog, recurrent deleted copy, unresolved and probable artifact.

The evidence differs between the classes of divergent sequence (Table 5). For the short-5.8S sequences of *Rhizoglomus*, all lines of evidence agree: all 40 share their deletion with sequences of other species, 22 recur in whole ITS under another culture label, 10 of the 10 from cultures sequenced in two or more libraries recur in another library of the same culture, and 26 match a public record at 99% identity or more. The 3 short-5.8S sequences of *Septoglomus viscosum* come from 2 cultures of one species and share a deletion of their own; 2 recur in a second library and match public records, so they are genomic, but whether they form a paralog beyond

this species is unknown, and their SSU is degraded in all three. The same-length sequences of *Gigaspora rosea* recur, identical, in other cultures and collections and are best read as co-dominant types of one genome. The other 5 same-length sequences, in *Septoglomus* and one in *Rhizoglomus intraradices*, are single sequences of unresolved origin. Only 16 of the 51 divergent sequences come from cultures with replicate libraries. The decision steps recognized the short-5.8S sequences of *Rhizoglomus* from their own sequence in 36 of 40 cases with their culture withheld and 29 with their species withheld, but none of the *Septoglomus viscosum* sequences in either test. Thus, the paralog interpretation is well supported only for *Rhizoglomus*, and none of these data shows whether a copy is transcribed.

Table 5: Evidence on the origin of each class of divergent sequence.

| | Short 5.8S *Rhizoglomus* | Short 5.8S *S. viscosum* | Same length *G. rosea* | Same length others |
|---|---|---|---|---|
| Sequences (cultures) | 40 (22) | 3 (2) | 3 (1) | 5 (4) |
| Named species | 5 + sp. | 1 | 1 | 2 + sp. |
| 5.8S deletion shared with another label | 40 | 3 | – | – |
| with another species | 40 | 0 | – | – |
| Recognized: culture / species withheld | 36 / 29 | 0 / 0 | – | – |
| Whole ITS recurs under another label | 22 | 0 | 3 | 0 |
| Same type in another library | 10 of 10 | 2 of 2 | 2 of 3 | 0 of 1 |
| Public record (≥99%) | 26 | 2 | 3 | 1 |
| Degraded structure (SSU / 5.8S / LSU) | 0 / 2 / 12 | 3 / 0 / 0 | 0 / 0 / 0 | 0 / 0 / 0 |
| Reading | paralog family | genomic | co-dominant types | unresolved |

*Shared deletion*: a 5.8S deletion of the same length, starting within 2 nt, in a sequence under another culture label (read after the check). *Whole ITS recurs*: a sequence within 1.0% in full ITS under another label, the criterion of the artifact check. *Recognized*: assigned to a divergent sequence by the decision steps in the tests with withheld cultures and species (Table 7); same-length sequences are only flagged. *Another library*: a sequence of the same type from an independent library of the same culture, among sequences whose culture has two or more libraries. *Public record*: a UNITE+INSD record at ≥99% identity over ≥90% coverage, culture sequences excluded. *Degraded*: broken base pairs above the 99th percentile of main-type sequences in the SSU, 5.8S or LSU (Methods S8). The same-length sequence of *Rhizoglomus intraradices* is counted with the others. None of these lines shows that a copy is transcribed.

The SSU of the short-5.8S sequences kept its secondary structure: only 3 of the 43 sequences were degraded, and their share of broken base pairs did not differ from that of the main-type sequences of their culture (median 10.0% against 10.6%; paired Wilcoxon test, $P = 0.17$). However, their LSU carries far more substitutions than the main-type sequences of their culture (median 4.62% against 0.12% of aligned positions), but most of these keep the base pairing: 12 sequences (28%) were degraded, and the share of broken base pairs differed little (7.6% against 8.5%; $P = 0.07$). The 5.8S differs most from that of the main-type sequences of the same culture: 25 of the 43 sequences were irregular, and broken base pairs rose from 4.0% to 16.0% (median paired difference +12.0 percentage points; $P < 10^{-8}$). In all 40 short-5.8S sequences of *Rhizoglomus*, the deletion removes three of the nine base pairs of a hairpin near the 3′ end of the 5.8S (positions 121–130 of the Rfam consensus structure), whereas in the 3 *Septoglomus viscosum* sequences it removes only positions that are unpaired in that structure. Under the rule fixed in advance, the short-5.8S class is therefore structurally degraded, as pseudogenes would be. The hairpin is shortened rather than destroyed, however: the remaining six-pair stem is made only of Watson–Crick pairs in 36 of the 40 sequences, and its new loop is UUCG in 25, UUUG in 10 and CUCG in 5; a UUCG loop closed by a C–G pair, as in most of these sequences, forms an unusually stable hairpin [50]. Two main-type sequences carry the same shortened hairpin, PX214776.1 (above) and PX214775.1 (*Glomus* sp. 5089_2351A), and both were among the sequences flagged by the artifact check. The same-length sequences were structurally intact: none was degraded in any region. In the ITS2, the predicted folds of short-5.8S sequences differed

from those of their culture's main type more than main-type sequences differ from each other (median base-pair distance 0.16 against 0.00 per nucleotide), with similar folding energies ($-0.23$ against $-0.26$ kcal mol$^{-1}$ per nucleotide). Thus, the 5.8S of the short type has lost part of one hairpin but keeps a stable, shortened stem.

Most short-5.8S sequences of *Rhizoglomus* do not group with the main type of their own culture (Figure 3): 39 of the 40 fall in 4 clades made only of short-5.8S sequences, 3 of which join two taxa (36 sequences). The groups of short-5.8S sequences, with SH-aLRT and ultrafast bootstrap support, include *Rhizoglomus irregulare* with *Rhizoglomus vesiculiferum* (26 sequences and 1 sequence classified as main type; 100/92); *Rhizoglomus intraradices* with *Rhizoglomus proliferum* (8 sequences; 100/99); *Rhizoglomus silesianum* (3 sequences; 100/100); *Rhizoglomus irregulare* with *Rhizoglomus* sp. (2 sequences; 100/100); *Rhizoglomus irregulare* (1 sequence). The short-5.8S sequences of *Rhizoglomus intraradices* and *Rhizoglomus proliferum*, two species that do not form a complex, are closer to each other than to the main type of either species. The short type is therefore most simply explained as a paralog that arose before these species separated and has been kept in their genomes since.

Counted by their 5.8S sequence, the Glomerales hold 44 short-5.8S sequences: the 43 of the divergent class and PX214776.1, which carries the same 5.8S but was classified as main type (see above). In the SSU region they show no history of their own: 21 are identical to a main-type sequence of their species, and all but one of the rest, a sequence of *Rhizoglomus* sp., sit next to main-type sequences of their own species. In the full ITS, 43 fall in clades made only of short-5.8S sequences, 3 of which join two taxa (37 sequences). In the LSU region, 42 do, but most of these clades hold one species, and 2 join two taxa (10 sequences). The markers then disagree: the 8 short-5.8S sequences of *Rhizoglomus intraradices* and *Rhizoglomus proliferum* (99/100) are sister to *Sclerocystis sinuosa* on the LSU tree (100/100), but to the main type of their own species on the ITS and concatenated trees (Supplementary Data S1). Thus, the paralog is visible in the ITS and LSU but not in the SSU.

Single-linkage clusters of all sequences of the 37 named species at 98.5% identity in the full ITS number 240 (Table S7). Main-type sequences alone give 220, so most of the excess over the number of taxa comes from natural intragenomic variation. The divergent sequences add 18 clusters made only of short-5.8S sequences and 2 made only of same-length sequences. Each of the 6 named species with the short type gains 1 to 7 such clusters, none of them shared with another species. On the Krüger fragment at 97%, the count falls to 56 (46 from main-type sequences), 8 of them made only of short-5.8S sequences. Thus, the short type raises species counts based on ITS clusters.

The shortened hairpin has a sequence of its own. In the main type, the region reads CACTCTCTGG(T/C)AACCCGGGGAGTATG, and in the short type CACTCTCTT(T/C) GGAGAGTATG. The 13-nt motif YTYGGRGARTATG, written from these sequences, occurs in the 5.8S of all 41 short-5.8S sequences of *Rhizoglomus* but in none of their other 113 sequences, and never outside the 5.8S in any culture sequence. It does not occur in the 3 short-5.8S sequences of *Septoglomus viscosum*, whose deletion lies at other positions; but it is the normal sequence of all 29 sequences of *Oehlia diaphana*, and it also occurs in PX214775.1, the main-type sequence of *Glomus* sp. 5089_2351A with the shortened hairpin. It starts at position 121–122 of the 5.8S, downstream of the binding sites of the ITS2 primers fITS7 and ITS3 (from positions 84–85 and 31), so ITS2 amplicons made with either primer include it (Methods S8). In UNITE, which holds none of the culture sequences, the motif splits the SHs of *Rhizoglomus*. Of the 1,355 sequences labelled *Rhizoglomus* or *Rhizophagus* that span the site (leaving out *Rhizophagus diaphanus*, the former name of *Oehlia diaphana*), 259 (19%) carry it. Of the 521 SHs in which they fall, 111 hold only such sequences (32 of them with two or more sequences), 407 none and 3 both. Outside these genera, the motif occurs in the 5.8S of 984 sequences, 940 of them in SHs with no sequence labelled *Rhizoglomus* or *Rhizophagus*, mostly unplaced Glomeraceae and *Glomus* sp. Thus, a short motif marks the paralog within *Rhizoglomus*.

On the trees, cultures were rarely monophyletic: 12%–20% of the 139 cultures with two or more sequences, depending on the region and on whether divergent sequences were kept (Table 6). A culture that is not monophyletic typically has most of its sequences outside any clade made only of its own sequences (median 67% on the concatenated sequence, main type), mixed with sequences of other cultures of the same species (Figure 3), as expected if the rRNA gene variants of a genome are older than the isolates that carry them. Named species mostly formed clades: 28, 24 and 28 of the 37 species in the SSU region, full ITS and LSU region, and 31 on the concatenated sequence. Removing the divergent sequences made 4 more species monophyletic on the concatenated sequence (*Rhizoglomus intraradices*, *Rhizoglomus irregulare*, *Rhizoglomus proliferum* and *Rhizoglomus vesiculiferum*), 35 of 37 in all. The two that remain, *Diversispora aestuarii* and *Glomus rugosae* (only the latter strongly), belong to species complexes, and all four complexes are monophyletic as units (4/4). Thus, species formed clades once divergent sequences were set aside, whereas conspecific cultures did not.

Table 6: Cultures, named species and species complexes that form a single clade (are monophyletic) on trees of the culture sequences.

| | Cultures | | Named species | | Complexes | |
|---|---|---|---|---|---|---|
| Tree | all | main | all | main | all | main |
| SSU region | 17/139 | 19/138 | 28/37 | 28/37 | 3/4 | 4/4 |
| Full ITS | 18/139 | 24/138 | 24/37 | 29/37 | 2/4 | 3/4 |
| LSU region | 19/139 | 22/138 | 28/37 | 30/37 | 3/4 | 4/4 |
| Concatenated sequence | 23/139 | 28/138 | 31/37 | 35/37 | 3/4 | 4/4 |

Monophyletic groups / groups tested; a group is tested when it has two or more sequences. *all*: all sequences; *main*: the same trees with the divergent sequences removed. One tree per order and region (maximum likelihood, IQ-TREE); the concatenated sequence joins the three regions. Alignments and trees are in Supplementary Data S1.

### 3.4 No single identity threshold separates species

A main-type sequence lies about as close to its nearest culture-mate as to the nearest sequence of another culture of its species (medians 0.41% and 0.66% in the full ITS, 0.14% and 0.14% in the LSU; Figure S1). Congeners overlap this range in every region: 39.5% of main-type sequences have a sequence of another species of their genus within the within-species limit of the full ITS, 54.5% within that of the LSU and 80.1% within that of the SSU region. Other genera stay outside it in every region except the NS31–AM1 fragment (13.5% of sequences). The database thresholds fall at different points on this scale. On the NS31–AM1 fragment, 97% identity, the VT threshold, joins 98.7% of main-type sequences with a congener and 15.2% with another genus (121 sequences, all but 4 in *Cetraspora*, *Gigaspora* and *Dentiscutata*). In the full ITS, the SH threshold of 98.5% identity lies well within the within-species limit: 24.4% of main-type sequences are more than 1.5% from every other main-type sequence of their own culture, which helps explain why SHs split cultures. In the LSU, the 97% threshold of V18 coincides with the within-species limit (3.0%).

Between pairs of cultures (Figure S2), the SSU distance was a median of 0.14 times the ITS distance (pairs at least 2% apart in ITS). Only 6 of the 5,196 pairs lay above half the ITS distance, all of them cultures of one species with SSU distances of at most 0.48%, so no pair of cultures met the SSU-discordance test (SSU at least 1% and at least half the ITS distance). That the test spares real sequences says nothing of what it catches, so we also gave it sequences of known origin (Methods S8). It flagged none of the 852 main-type sequences scored with their culture withheld, although 26 of them lie at least 1% from every other culture in the SSU

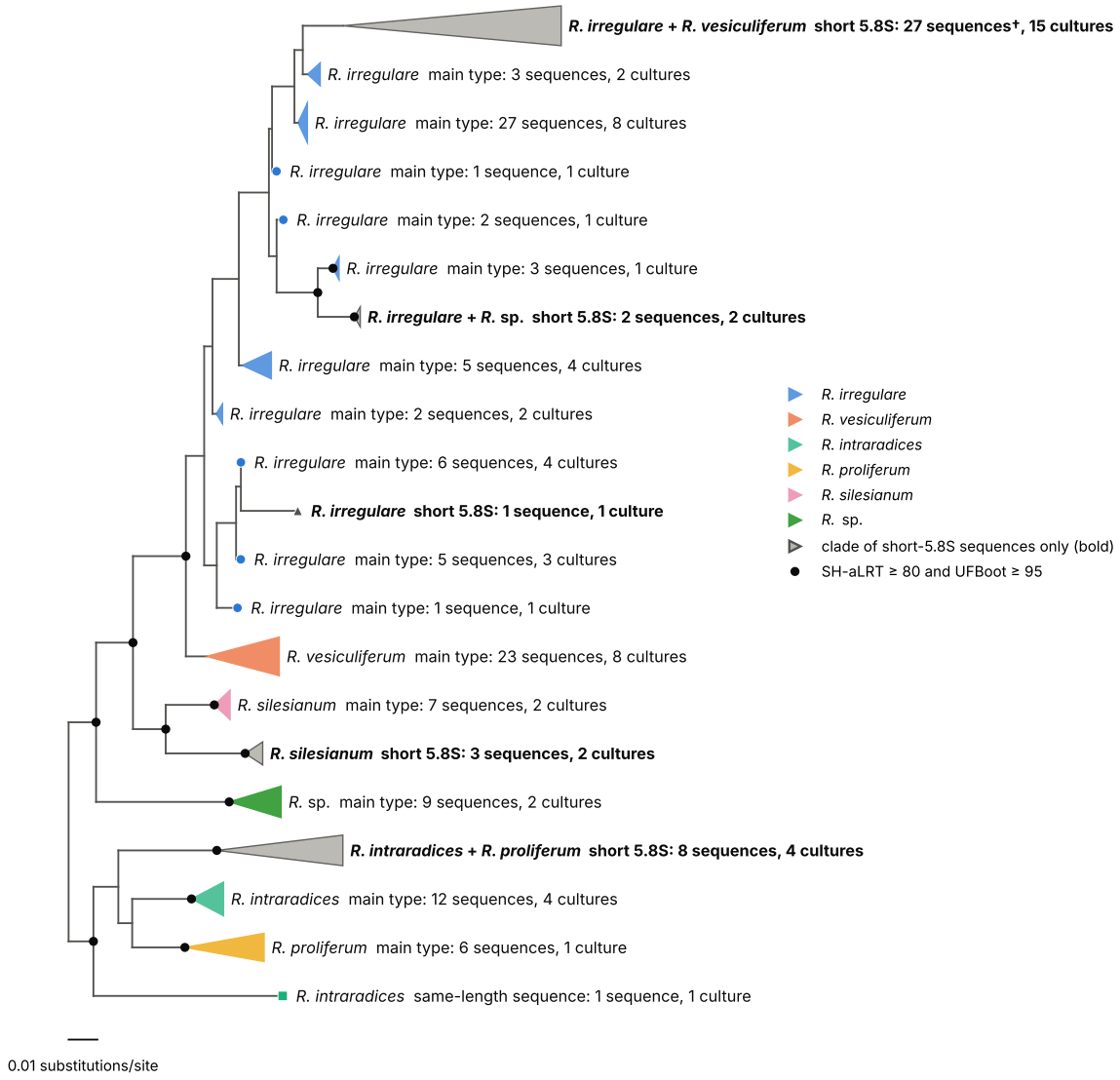



Figure 3: Phylogeny of *Rhizoglomus* showing that short-5.8S sequences group across species. The *Rhizoglomus* clade of the maximum-likelihood tree of the concatenated sequences (SSU region, full ITS and LSU region) of the Glomerales, rooted on *Funneliformis* (not shown). Clades made only of main-type sequences of one species (coloured), and clades made only of short-5.8S sequences (grey, bold labels), are collapsed into triangles whose length is the depth of the clade; labels give the species (*R.*, *Rhizoglomus*), the sequence class and the numbers of sequences and cultures. Black dots mark branches with SH-aLRT $\geq 80$ and ultrafast bootstrap $\geq 95$. †One of the 27 sequences was classified as main type (PX214776.1, *Rhizoglomus irregulare* 4569_1185B) but carries the deletion of the short-5.8S sequences. The uncollapsed tree (154 sequences, 103 distinct sequences) is Figure S3, and all trees are in Supplementary Data S1.

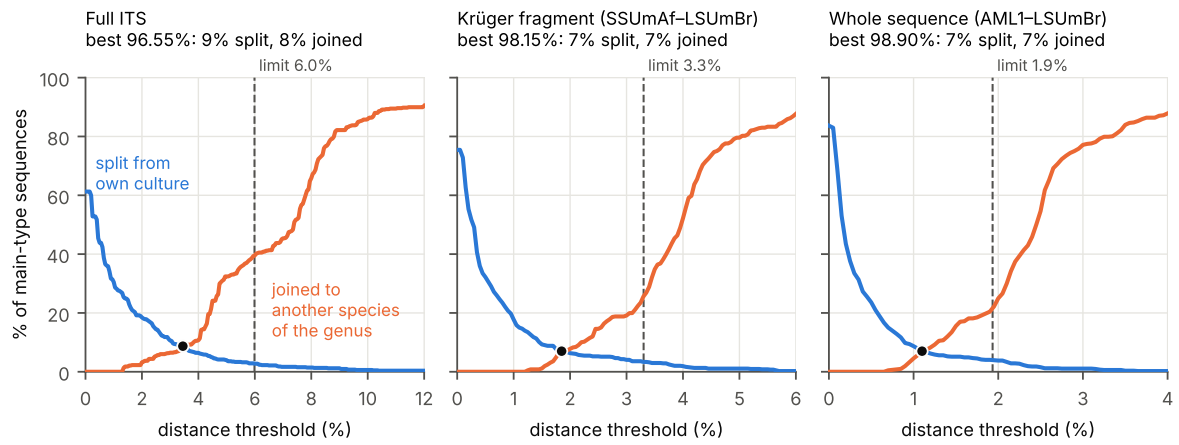



Figure 4: Trade-off between splitting genomes and joining species at single identity cut-offs on three regions. For each cut-off, the share of main-type sequences more than the cut-off from every other sequence of their culture (blue) and with a sequence of another species of their genus within the cut-off (orange), on the full ITS, the Krüger fragment (SSUmAf site to LSUmBr) and the whole sequence (AML1 to LSUmBr). Dashed lines, the within-species limit; dots, the cut-off that minimizes the larger of the two shares, given above each panel. The four complexes count as separate species here; leaving their partners out, the share joined at the within-species limit falls to 11.8% (full ITS), 2.1% (Krüger fragment) and 1.9% (whole sequence; Methods S8).

region. Errors spread along a sequence were flagged once they reached a few percent: 0.7% of 148 sequences with 0.5% of positions changed, 29% with 2% and 92% with 8%. Chimeras were almost never flagged: none of the 300 that join the SSU of one species to the ITS of another, because their SSU is close to that of its parent, and 10 of the 300 that switch parents inside the SSU, because their nearest SSU usually belongs to a lineage whose ITS is even more divergent. Thus, the first decision step flags SSU discordance and is sensitive to reads with many errors, but it is not a chimera screen.

Taken together, the regions of the Krüger fragment (from the SSUmAf site to LSUmBr; median 1,577 nt) keep genomes and species apart better than the full ITS alone (Figure 4). At 97% identity, 4.1% of main-type sequences lie more than 3% from every other sequence of their culture, against 24.4% for the full ITS at 98.5%, and outside the four complexes 0.3% have a congener within 3%. The complexes stay joined at any cut-off that keeps genomes together: counted as separate species, they leave the best single cut-off (98.15%) splitting 6.7% of sequences from their culture and joining 7.0% to another species. The whole sequence (median 2,758 nt) behaves the same way at a higher identity (best cut-off 98.90%: 7.0% split, 7.0% joined). Thus, no single identity threshold separates the species, although longer regions come closer than the ITS.

### 3.5 Combining the three naming systems to identify species

Combining the names recovered what no single system reaches. With a culture withheld, the exact species of the 755 main-type sequences of species held by another culture came from the VT alone for 370 (49%), from the SH alone for 343 (45%), from the LSU alone for 351 (46%) and from the consensus assignment for 607 (80%); the checks for new sequences then changed 4 of these to 'new', leaving 603 (Table 7). Placing the LSU on the reference tree raised exact assignments to 701 and cut false-known assignments for withheld species from 26.8% to 8.8%, and the ITS distance rule cut them again to 2.2% (Figure 5a). Averaged over species or over cultures, the final rate was 1.4% and 1.9%, respectively; its 95% interval was 0.2–5.0% when

species were resampled and 1.0–3.7% when cultures were resampled, with errors in 4 species and 11 cultures (Table 12). The limit of the ITS distance rule was 6.0% in 139 of 148 rounds. The rule changed 6 sequences of known species to 'new' (0.8%; 0.0–2.1% with cultures resampled; 4 from *Glomus chinense* 4036_906B, 1 from *Gigaspora rosea* CU131_1018E, 1 from *Microdominikia litorea* 5274); the *Microdominikia litorea* sequence lies 12.6% from the species' other culture. The 18 remaining false-known assignments belong to two complexes: *Diversispora varaderana* with *Diversispora aestuarii* (9 sequences), which no marker separates, and *Glomus chinense* with *Glomus rugosae* (9 sequences), whose ITS ranges overlap.

Of the 42 divergent sequences whose species is held by another culture, 35 were recognized as divergent, 3 same-length sequences were assigned to their own species, and 4 were wrongly considered a new species. Short-5.8S sequences were recognized from their own sequence in 36 of 43 cases with their culture withheld, and in 29 with their species withheld, all of them in *Rhizoglomus* (Table 5). Placement has a cost: 146 withheld sequences changed from 'new species in a known genus' to 'new lineage', which has no genus.

Table 7: Assignment of the sequences of withheld cultures and species, with the external databases intact.

| LSU naming | Known kept (of 755) | Exact species | False known species withheld (of 803) | False known single-culture species (of 48) |
|---|---|---|---|---|
| Nearest LSU reference | 748 | 603 | 215 (26.8%) | 3 |
| LSU species clade (placement) | 751 | 701 | 71 (8.8%) | 0 |
| Placement + ITS distance rule | 745 | 697 | 18 (2.2%) | 0 |

*Known kept*: main-type sequences of species represented by another remaining culture that were assigned to a known species or species group that includes their own species (148 rounds, one culture withheld in each). *False known*: sequences assigned to a known species although their species has no remaining culture, in 50 rounds with one species withheld in each and, in the rounds with one culture withheld, among the main-type sequences of species with a single culture. Averaged over species and over cultures, respectively, the final false-known rate with a species withheld is 1.4% and 1.9%.

### 3.6 Removing the withheld species from the LSU reference

With the full LSU reference, sequences of *Glomus rugosae* fall in the reference's own *Glomus rugosae* clade, which no remaining culture holds, and are flagged as new. With the species removed from the reference, they fall in the *Glomus chinense* and *Glomus macrocarpum* clades and are assigned to known species (Table 8; Figure 5b). The ITS distance rule does not use the LSU reference, but it rescued only 13 of the 75 sequences, because the ITS of *Glomus rugosae* lies 1.4–7.5% (median 4.7%) from its relatives, within the ordinary variation of one species.

Pooled over the 34 test species (673 sequences), false-known assignments numbered 19 with the LSU reference intact, 78 with the species removed from it, and 53 with the ITS distance rule as well (Figure 5c; per species in Table S4); averaged over species, the rates were 20.6% and 11.7%, respectively. With the ITS rule, the pooled rate of 7.9% has a 95% interval of 1.4–17.3% when species are resampled (3.5–13.1% when cultures are), and the errors come from 6 of the 34 species (Table 12). This is the most realistic of our tests of an absent species, and its rate, not the 2.2% of the test with the databases intact, is the better guide for an absent species with close relatives in the reference. The 53 errors left after the ITS rule fall in 6 species pairs: *Diversispora aestuarii* assigned to *Diversispora varaderana* (20), *Rhizoglomus irregulare* assigned to *Rhizoglomus vesiculiferum* (18), *Diversispora arenaria* assigned to *Diversispora jakucsiae* (4), *Diversispora jakucsiae* assigned to *Diversispora arenaria* (4), *Glomus macrocarpum* assigned to *Glomus rugosae* (4), *Diversispora varaderana* assigned to *Diversispora aestuarii* (3). Of the 673 withheld sequences, 272 were considered a new species in a known genus (195 because the LSU

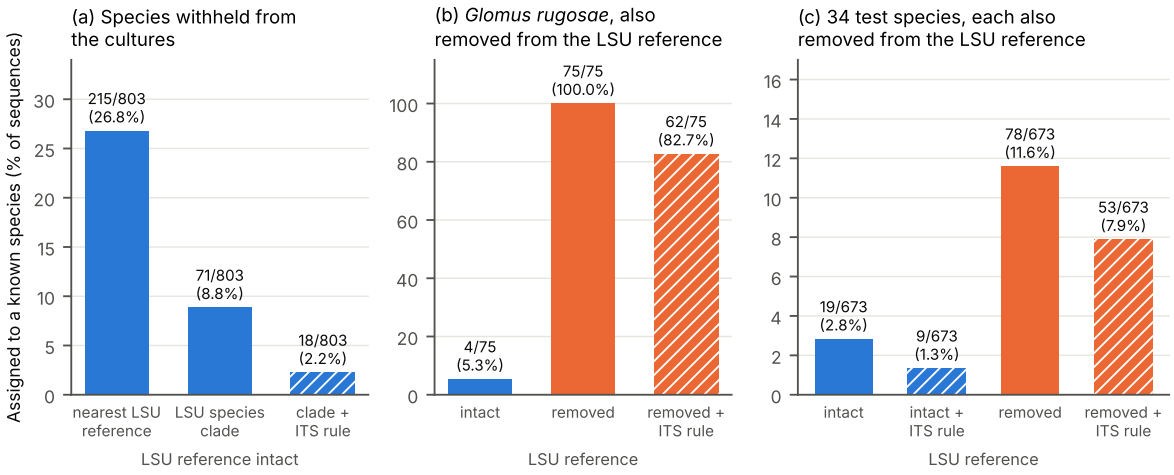



Figure 5: False-known assignments, i.e., sequences assigned to a known species although no remaining culture holds their species. (a) Species withheld from the cultures, with the LSU reference intact, by how the LSU was used. (b) *Glomus rugosae*, a development species, also removed from the LSU reference: the signal of novelty that LSU placement gave came from the LSU reference itself. (c) The 34 test species, each also removed from the LSU reference, pooled. Blue, LSU reference intact; orange, species removed from it; hatched, with the ITS distance rule.

name was unmatched, 34 because the names shared no species, 25 by the ITS distance rule, and 18 because the VT was unmatched), 347 a new lineage, 53 a known species or species group, and 1 a divergent sequence. A genus was given to 326 sequences and was correct for 326 of them, but that measures precision only. Of the 487 sequences whose genus another culture holds, the genus was kept for 326 (66.9%), and 161 were classified as a new lineage although their genus was known; the other 186 sequences belong to genera with no other culture and were all classified as a new lineage. The VT thus contributes little to the detection of new species (2.7% of sequences), and only when the withheld species has a VT of its own. *Ambispora callosa*, which the LSU reference does not hold by name, was assigned to a known species for 20/20 sequences without the ITS rule and 0/20 with it, because its ITS lies about 10% from its closest relative.

Table 8: False-known assignments when the withheld species were also removed from the LSU reference.

| Withheld species | LSU reference | False known | with ITS rule |
|---|---|---|---|
| *Glomus chinense*† | intact | 28/35 | 5/35 |
| *Glomus chinense*† | species removed | 28/35 | 5/35 |
| *Glomus rugosae*† | intact | 4/75 | 4/75 |
| *Glomus rugosae*† | species removed | 75/75 | 62/75 |
| 34 test species | intact | 19/673 (2.8%) | 9/673 (1.3%) |
| 34 test species | species removed | 78/673 (11.6%) | 53/673 (7.9%) |

Numbers of main-type sequences of each withheld species that were assigned to a known species. *Intact*: sequences placed on the full LSU reference (V18); *species removed*: V18 realigned without the species' records, the model refitted and all sequences placed again. † Development species, used to develop the ITS distance rule. The test species, i.e., all other species whose V18 records could be removed, are pooled; *Ambispora callosa*, which V18 does not hold by label, is reported apart.

With a whole genus withheld from the cultures and from the LSU reference (15 rounds, 803

main-type sequences of named species), no sequence was assigned to a known species, with or without the ITS rule, as expected, because no remaining culture shares the genus. Of these sequences, 734 were classified as a new lineage, the correct rank, and the other 69 (8.6%) as a new species in a known genus, and that genus was wrong in every case: *Rhizoglomus* assigned to *Sclerocystis* (69). With the LSU reference intact, 0 sequences received a wrong genus. This failure is easy to miss in practice, because these sequences receive a definite genus rather than a flag. The genus test shows at the level of genera what the species test shows at the level of species: a sequence whose clade has been removed is placed on the sister clade, here *Sclerocystis*, and named after the cultures that hold it. None of the 51 divergent sequences was recognized as divergent in these rounds, because both divergent-sequence tests need a related culture, and 40 of them also received the wrong genus. Of the diagnostics for the short type, only the motif of *Rhizoglomus* (Section 3.3) does not depend on a related culture, and it was not part of the fixed steps.

### 3.7 The remaining errors and the species complexes

The complex rule yields 4 complexes (Table 9): *Glomus chinense*, *Glomus macrocarpum* and *Glomus rugosae* (the VTX00199 group); *Diversispora aestuarii* with *Diversispora varaderana*; *Diversispora arenaria* with *Diversispora jakucsiae*; and *Rhizoglomus irregulare* with *Rhizoglomus vesiculiferum*. Together they hold 9 of the 37 named species and 29.9% of the reference sequences. One further pair, *Diversispora celata* with *Diversispora eburnea*, misses the LSU limit by 0.4 percentage points and is not a complex.

Every one of the 53 false-known assignments left after the ITS rule in the test species is an assignment to a member of the species' own complex. Without the ITS rule, 54 of the 78 errors are inside a complex and 24 (3.6% of the withheld sequences) are not; with the LSU reference intact, 10 and 0 sequences fall outside a complex without and with the rule, respectively. The ITS rule thus removes the errors that fall outside a complex. This is a retrospective statement: the complexes were derived from all cultures, the withheld species included, so each species under test helped define the groups against which its own errors are judged. It shows that the errors observed here lie in pairs that are inseparable when all cultured species are considered, not that the rule could have predicted them.

Table 9: Pairs of named species that form species complexes, i.e., whose closest sequences lie within the within-species limit on all three markers.

| Species A | Species B | Complex | ITS | LSU | SSU |
|---|---|---|---|---|---|
| *Diversispora aestuarii* | *Diversispora varaderana* | C2 | 1.1 | 1.09 | 0.0 |
| *Diversispora arenaria* | *Diversispora jakucsiae* | C3 | 3.48 | 0.83 | 0.07 |
| *Glomus chinense* | *Glomus macrocarpum* | C1 | 5.87 | 2.18 | 0.27 |
| *Glomus chinense* | *Glomus rugosae* | C1 | 4.95 | 1.36 | 0.2 |
| *Glomus macrocarpum* | *Glomus rugosae* | C1 | 1.38 | 1.36 | 0.07 |
| *Rhizoglomus irregulare* | *Rhizoglomus vesiculiferum* | C4 | 3.25 | 2.44 | 0.34 |

Columns give the distance (%) between the nearest sequences of the two species. The limits are the 99th percentile of the distance between cultures of one species: ITS 6.0%, LSU 3.0%, SSU 0.8%.

To ask whether the rule could have predicted them, we applied it as it would be applied to a species that no culture holds (Table 10). In each round, the withheld cultures were removed and the within-species limits recomputed without them (5.98–6.08% in the ITS, 2.31–3.11% in the LSU and 0.74–0.81% in the SSU region across rounds), and each sequence of the withheld species was compared with every remaining species. 81 of the 673 sequences of the test species lay inside a remaining species on all three markers, in 7 species. At the level of species, the rule anticipated the errors: each of the 6 species with false-known assignments after the ITS rule had

sequences inside a congener, none of the 27 species without such sequences had a false-known assignment, and one species predicted to be at risk, *Rhizoglomus vesiculiferum*, had none. At the level of sequences, it fell short. Only 13 of the 56 sequences of *Rhizoglomus irregulare* lie inside *Rhizoglomus vesiculiferum*; the others lie outside it in LR0R–FLR2 (42 sequences, 3.5–5.4% against a recomputed limit of 2.95%) or, in 1 case, in the ITS alone; yet 18 were assigned to *Rhizoglomus vesiculiferum*. Matched sequence by sequence, 5 of the 53 errors lie outside the prediction, and the other 48 were each assigned to a species they lie inside; without the ITS rule, 30 of 78 lie outside. The errors outside the prediction with the rule all involve the one pair whose status as a complex depends on the percentile (below).

Table 10: The complex rule applied as a prediction within each round of the LSU-removal test.

| Withheld species | Sequences | Complex (all cultures) | Inside a remaining congener $n$ | epithet ($n$) | False known no rule | False known ITS rule |
|---|---|---|---|---|---|---|
| *Ambispora leptoticha* | 10 | – | 0 | – | 10 | 0 |
| *Diversispora aestuarii* | 20 | C2 | 20 | *varaderana* (20) | 20 | 20 |
| *Diversispora arenaria* | 4 | C3 | 4 | *jakucsiae* (4) | 4 | 4 |
| *Diversispora jakucsiae* | 4 | C3 | 4 | *arenaria* (4) | 4 | 4 |
| *Diversispora varaderana* | 31 | C2 | 31 | *aestuarii* (31) | 3 | 3 |
| *Funneliformis geosporum* | 8 | – | 0 | – | 8 | 0 |
| *Glomus macrocarpum* | 7 | C1 | 7 | *rugosae* (7); *chinense* (1) | 4 | 4 |
| *Rhizoglomus irregulare* | 56 | C4 | 13 | *vesiculiferum* (13) | 19 | 18 |
| *Septoglomus africanum* | 6 | – | 0 | – | 6 | 0 |
| *Rhizoglomus vesiculiferum* | 23 | C4 | 2 | *irregulare* (2) | 0 | 0 |
| *Glomus chinense*† | 35 | C1 | 6 | *rugosae* (6) | 28 | 5 |
| *Glomus rugosae*† | 75 | C1 | 62 | *macrocarpum* (62); *chinense* (9) | 75 | 62 |
| 24 other test species | 504 | – | 0 | – | 0 | 0 |

Main-type sequences of each withheld species; $n$, number of sequences. *Complex*: the species' complex when the rule is computed from all cultures (Table 9). *Inside*: with the withheld cultures removed and the within-species limit recomputed without them, sequences that lie within the limit of a remaining species on all three markers; only these can be assigned to a known species if the rule holds. False-known assignments with the species removed from the LSU reference (Table S4). † Development species.

The complexes depend on the percentile. *Diversispora aestuarii* with *Diversispora varaderana*, *Diversispora arenaria* with *Diversispora jakucsiae*, and *Glomus macrocarpum* with *Glomus rugosae* are complexes at every percentile from the 95th to the 99.5th; *Glomus chinense* joins from the 97.5th, *Rhizoglomus irregulare* with *Rhizoglomus vesiculiferum* from the 99th, and *Funneliformis coronatus* with *Funneliformis mosseae* at the 99.5th. Because the ITS distance rule uses the same percentile, a fair test changes both together (Table S5). At the 97.5th percentile, the rule leaves the same 53 false-known assignments as at the 99th, but the 18 sequences of *Rhizoglomus irregulare* assigned to *Rhizoglomus vesiculiferum* now lie outside a complex. At the 95th percentile the rule is stricter: false-known assignments fall to 37 (5.5%), 5 of them outside a complex, all again *Rhizoglomus irregulare* assigned to *Rhizoglomus vesiculiferum*, at the cost of changing 32 sequences of known species to 'new' in the culture test (4.2%) instead of 6. At every percentile, then, each remaining error is either inside a complex or in this one pair, and whether that pair counts as a complex is the only conclusion that depends on the choice.

### 3.8 Cultures from other collections

Before any test result was scored, the 850 sequences of the 37 LSU-removal rounds were scored again with the test cultures present but excluded, and every assignment came back unchanged; adding the test ASVs to the LSU alignment left the 903 reference placements unchanged.

Among the 102 ASVs of species absent from the reference, 4 were assigned to a known species with the species removed from the LSU reference (3.9%; 5.3% averaged over species and 3.8% over cultures; 95% interval 0.0–13.2% with cultures resampled; Table 11). These are exactly

the 4 ASVs that the prediction had placed inside a reference species (*Gigaspora gigantea* KS302 assigned to *Gigaspora rosea*); no false-known assignment fell outside the prediction (Figure 6a). With the LSU reference intact, their LSU falls in its *Gigaspora gigantea* clade, which the reference's *Gigaspora rosea* cultures also occupy, and their ITS lies 0.5–2.0% from those cultures, so no marker separates the two. None of the 43 ASVs of genera absent from the reference was assigned to a known species. Removing the species from the LSU reference changed no assignment: all 102 ASVs of absent species had the same assignment and placement either way, because their LSU fell in clades that no reference culture holds in both cases. These species are distant from those of the reference, and the sister-species case behind the LSU-removal result (Section 3.6) occurred once, in KS302, where it was predicted.

Of the 164 ASVs of species present in the reference, 136 kept their species (82.9%), against 745 of 755 in the culture test. ASVs identical in ITS to a reference sequence kept it in 40 of 43 cases, the others in 96 of 121 (79.3%). The ITS distance rule changed 15 of them to 'new', because their ITS lies 6.1–9.6% from every reference culture, beyond its 6.0% limit: without the rule, 151 kept their species (92.1%), and the rule removed no false-known assignment (4 without it). The rule thus changed 9.1% of these ASVs (2.0–21.4% with cultures resampled), against 0.8% (0.0–2.1%) of known-species sequences in the culture test: its limit, fitted on one collection, did not carry over to another, and here it lowered retention without preventing an error. The 8 ASVs assigned to another species are *Funneliformis coronatus* MR101 assigned to *Funneliformis mosseae*, as predicted (8 of 8): their ITS lies 0.0–0.6% and their SSU 0.0% from reference *Funneliformis mosseae* cultures, so the culture is most likely *Funneliformis mosseae* under another name. The prediction also sorted the known species: 135 of the 139 ASVs predicted inside their own species kept it, and 24 of the 25 predicted outside did not.

With the VT search nearly silent, the genus of absent species was hard to identify: 44 of the 59 ASVs of absent species in genera present in the reference were classified as a new lineage, 9 were near a VT threshold, and the genus was given for 6, all correctly. With VT coverage counted over the part of the reference that the ASV spans, the genus was given for 29, again all correctly, but two errors appeared: the 18 ASVs of *Racocetra gregaria* were given *Cetraspora*, through a VT shared by the two genera (VTX00041), while their LSU name was unmatched; and one ASV, *Septoglomus constrictum* KS890 assigned to *Septoglomus africanum* / *Septoglomus turnauae*, through a shared VT, with its ITS 1.1% from a reference culture. False-known assignments then numbered 5 (4.9%), 1 of them outside the prediction.

The test also raised curation questions for the collections. Three culture codes carry different species in LongGloDB and in the Stefani et al. deposit (MT106, ON205A and BEG9). INVAM's name for both MT106 and ON205A is *Ambispora gerdemannii*, which Stefani et al. record and depart from (their Table S2) and which LongGloDB keeps for ON205A. In addition, the LongGloDB *Ambispora leptoticha* MX982A lies 7.8–9.6% in ITS from the reference's *Ambispora leptoticha*, ON205A, which LongGloDB names *Ambispora gerdemannii*.

### 3.9 Sequences from EUKARYOME

Unlike the LongGloDB amplicons, nearly all EUKARYOME sequences cover the VT fragment: 5,213 of the 5,511 had a VT hit that passed, whereas only 1,379 had a UNITE hit that passed. Adding the test sequences changed no result of the LSU-removal test (850 sequences in 37 rounds, rescored). Of the 2,706 sequences of absent taxa, 13 (0.5%; averaged over taxa, 2.9%) were assigned to a known species or species group, with each absent species or genus removed from the LSU reference where V18 holds it (Table 13; Figure 6c). All of them were sequences of absent species in genera present in the reference: 13 of 77 (16.9%), from 4 of the 33 species. No sequence of an absent genus or placeholder lineage was assigned to a known species. The count was 13 with the LSU reference intact as well, 14 with VT coverage counted over the spanned part, and 24 without the ITS distance rule. None of the false-known sequences was among those flagged before the run. Thus, the errors concentrated where the test is hardest, in close relatives

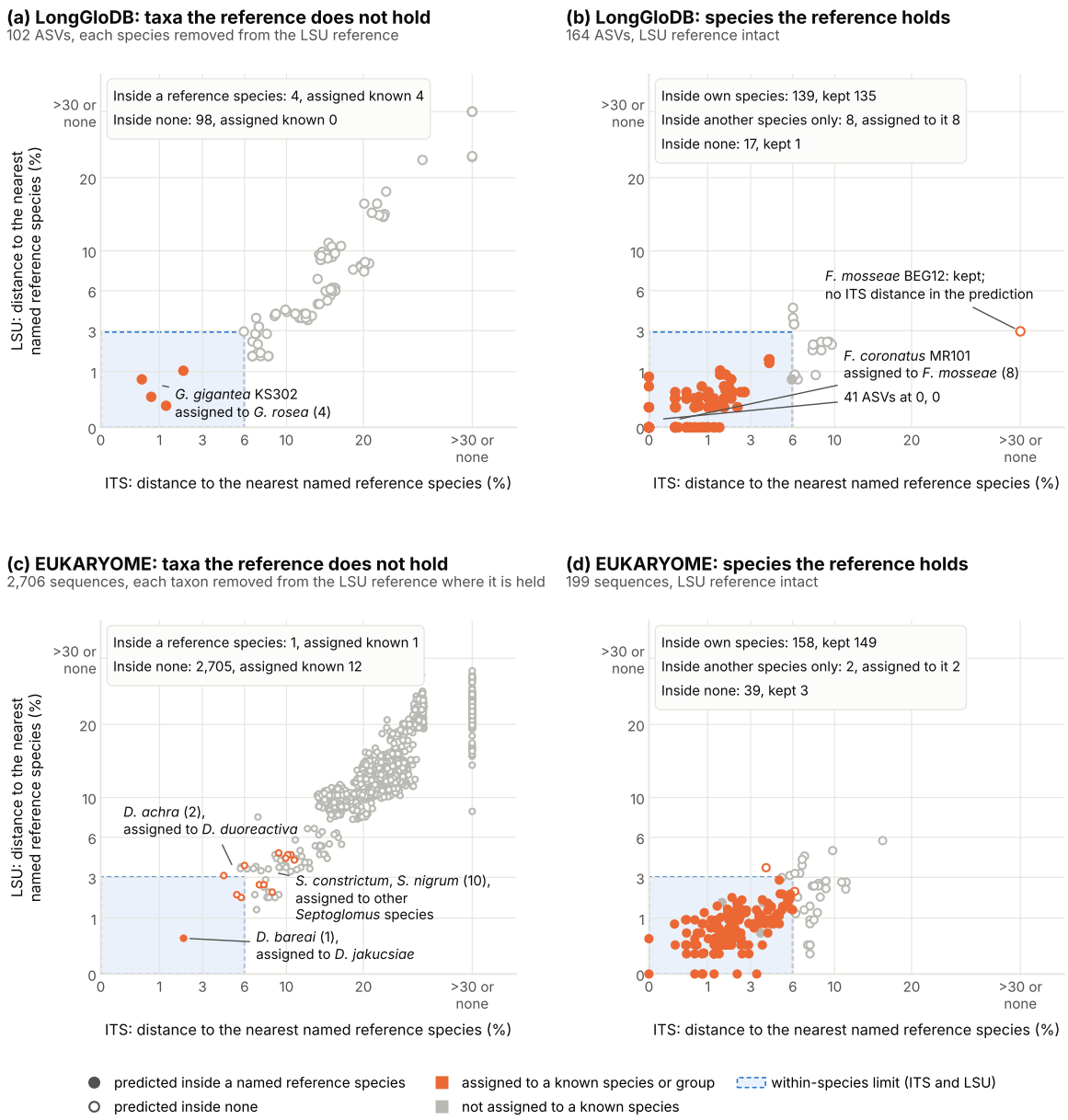



Figure 6: Predicted versus observed assignments in the two independent tests. Each point is one test sequence, a LongGloDB ASV or a EUKARYOME sequence, placed by its distance to the nearest named reference species in the full ITS and in LR0R–FLR2 (square-root axes; the nearest species can differ between the two regions). The shaded box is the within-species limit used in the locked predictions (Methods S7). Filled points were predicted, before the run, to lie inside a named reference species on all three markers, the SSU region included; orange points were assigned to a known species or species group. (a) The 102 ASVs of species and genera absent from the reference, each with its species removed from the LSU reference: the only known-species assignments are the 4 predicted ASVs of *Gigaspora gigantea* KS302 assigned to *Gigaspora rosea*. (b) The 164 ASVs of species present in the reference, with the LSU reference intact; *Funneliformis coronatus* MR101 lies inside *Funneliformis mosseae* and was assigned to it. (c) The 2,706 EUKARYOME sequences of absent taxa, each with its species or genus removed from the LSU reference where V18 holds it: one known-species assignment was predicted and 12 were not (Section 3.9). (d) The 199 EUKARYOME sequences of reference species, with the LSU reference intact. Points at the edge of an axis lie more than 30% from every named reference species, or lack that region in the files on which the predictions were computed.

Table 11: Independent cultures: the rules as fixed, on 52 LongGloDB cultures from other collections.

| Analysis | LSU reference | False known (of 102) | Share | Absent species: genus right/wrong/none (of 59) | Absent genera: genus given (of 43) | Known kept (of 164) |
|---|---|---|---|---|---|---|
| Main | intact | 4 (4) | 3.9% | 6 / 0 / 53 | 0 | 136 (151) |
| | removed | 4 (4) | 3.9% | 6 / 0 / 53 | 0 | – |
| Second | intact | 5 (5) | 4.9% | 29 / 0 / 30 | 18 | 139 (154) |
| | removed | 5 (5) | 4.9% | 29 / 0 / 30 | 18 | – |

With the ITS distance rule; in brackets, the same without it. *LSU reference*: intact, or with each absent species removed in its own round. *False known*: ASVs of species the reference does not hold assigned to a known species or species group. *Absent species*: ASVs of species the reference does not hold, in genera it holds. *Absent genera*: ASVs of genera it does not hold, so any genus given is wrong. *Known kept*: ASVs of species the reference holds whose assignment contains their species; LSU reference intact only. *Second*: the second analysis, fixed in advance, with VT coverage counted over the part of each MaarjAM reference that the ASV spans.

Table 12: Main error and retention rates, with 95% intervals that resample species or cultures.

| Test | Outcome | Events | 95% interval (units with events) resampling species | resampling cultures |
|---|---|---|---|---|
| Withheld cultures | kept | 751/755 (99.5%) | 98.8–99.9% (30/30) | 98.9–99.9% (131/131) |
| | kept, ITS rule | 745/755 (98.7%) | 97.2–99.6% (30/30) | 97.2–99.6% (131/131) |
| | demoted by ITS rule | 6/755 (0.8%) | 0.0–2.2% (3/30) | 0.0–2.1% (3/131) |
| Withheld species | false known | 71/803 (8.8%) | 1.6–20.3% (6/37) | 4.1–14.3% (17/138) |
| | false known, ITS rule | 18/803 (2.2%) | 0.2–5.0% (4/37) | 1.0–3.7% (11/138) |
| LSU-removal test | false known, intact, ITS rule | 9/673 (1.3%) | 0.0–4.3% (2/34) | 0.3–2.7% (6/122) |
| | false known, removed | 78/673 (11.6%) | 3.9–23.2% (9/34) | 6.3–17.9% (21/122) |
| | false known, removed, ITS rule | 53/673 (7.9%) | 1.4–17.3% (6/34) | 3.5–13.1% (15/122) |
| LongGloDB | false known, removed, ITS rule | 4/102 (3.9%) | 0.0–14.4% (1/19) | 0.0–13.2% (1/26) |
| | kept | 151/164 (92.1%) | 58.2–99.6% (6/9) | 78.7–100.0% (19/22) |
| | kept, ITS rule | 136/164 (82.9%) | 37.3–95.2% (6/9) | 66.0–94.5% (18/22) |
| | demoted by ITS rule | 15/164 (9.1%) | 0.8–33.8% (3/9) | 2.0–21.4% (6/22) |

*Events*: events over sequences (ASVs for LongGloDB); *kept*: assigned to a known species that contains their own; *intact*, *removed*: the LSU reference intact, or with the species removed from it. Pooled rates with percentile intervals from 10,000 cluster-bootstrap replicates that resample species or cultures with replacement. In brackets: species or cultures with at least one event, of those tested. Withheld cultures and species: external databases intact, LSU placement. LSU-removal test: the test species; intervals by culture need the per-sequence results and are not given. LongGloDB: known species with the LSU reference intact; absent species with their species removed from it.

of reference species.

The locked prediction placed one sequence of an absent taxon inside a reference species on all three markers: *Diversispora bareai* inside *Diversispora arenaria* and *Diversispora jakucsiae*. As predicted, it was assigned to *Diversispora jakucsiae*. The other 12 false-known assignments lay outside the prediction, so they contradict the claim of Section 3.7 for these sequences. 10 of them, labelled *Septoglomus constrictum* and *Septoglomus nigrum*, lie inside one of 2 reference cultures identified only to genus, *Septoglomus* sp. 4404_2057A and FS_9_2328B (ITS 1.3–5.6%, SSU 0.0–0.7%). The prediction, like the complex rule, measured distances to named species only. These sequences matched VTs that the genus-only cultures share with named species (VTX00064 with *Septoglomus africanum* and *Septoglomus turnauae*, VTX00063 with *Septoglomus viscosum*), and LSU species clades linked to no named species, so the consensus kept the species of the VT. The other 2 sequences, labelled *Dominikia achra*, lie within the ITS and SSU limits of *Dominikia duoreactiva* but 3.1–3.8% from it in the LSU, above the LSU limit; LSU placement nevertheless put them in its clade. Whether the EUKARYOME labels or the culture names are right cannot

be decided here (Section 2.8). Thus, the prediction missed errors that arose through cultures identified only to genus and through LSU placement near the limit.

Of the 199 sequences of reference species, 152 (76.4%) were assigned to their own species, alone or in a group, and 7 to another species. Here the ITS distance rule prevented 11 false-known assignments, but it also changed 22 sequences of reference species to 'new', so that 20 fewer were kept (172 without the rule). None of these 22 sequences lay inside its labelled species on all three markers; in the ITS, they lay 6.2–8.6% from the nearest culture of that species. A further 14 sequences, all labelled *Rhizoglomus irregulare* or *Rhizoglomus intraradices*, were classified as divergent sequences of the short-5.8S type.

For absent species in reference genera, the genus given was right for 49 of the 76 sequences with a genus outcome, wrong for 0, and missing for 27. For absent genera, any genus given is wrong. With each such genus removed from the LSU reference, 268 of 1,636 sequences (16.4%) were given a genus, most of them *Orientoglomus* (114 sequences, given *Microdominikia*) and *Silvaspora* (88 sequences, given *Sclerocystis*); with the LSU reference intact, 61 (3.7%). Thus, a genus missing from the LSU reference was often replaced by a related genus, as in the LSU-removal test.

The short-5.8S motif occurs in 95 EUKARYOME sequences labelled *Rhizoglomus*. Of these, 46 were classified as divergent sequences, but 28 were classified as new lineages and 20 as new species in a known genus. Thus, in field sequences, the decision steps recognized the paralog in about half of the cases, and the rest would have been counted as new taxa.

Table 13: Assignment of the 5,511 EUKARYOME sequences with the rules as fixed in advance.

| Analysis | LSU reference | False known (of 2,706) | Share | Absent species: genus right/wrong/none (of 76) | Absent genera: genus given (of 1,636) | Known kept (of 199) |
|---|---|---|---|---|---|---|
| Main | removed | 13 (24) | 0.5% | 49 / 0 / 27 | 268 | – |
| | intact | 13 (21) | 0.5% | 44 / 0 / 32 | 61 | 152 (172) |
| Second | removed | 14 (25) | 0.5% | 50 / 0 / 26 | 271 | – |
| | intact | 14 (22) | 0.5% | 45 / 0 / 31 | 64 | 152 (172) |

Counts with the ITS distance rule; in brackets, without it. *LSU reference*: *removed*, each absent species or genus that V18 holds was removed in its own round (sequences of taxa that V18 does not name were placed on the full reference); *intact*, every sequence was placed on the full reference. *False known*: sequences of taxa absent from the reference that were assigned to a known species or species group. *Absent species*: sequences of species absent from the reference, in genera present in it (*Dentiscutata erythropus*, whose genus is uncertain, not counted). *Absent genera*: sequences of genera absent from the reference and of EUKARYOME placeholder lineages, for which any genus given is wrong (*Nanoglomus* not counted; Section 2.8). *Known kept*: sequences of species present in the reference whose assignment includes their own species. *Second*: VT coverage counted over the part of each MaarjAM reference that the sequence spans.

# 4 Discussion

Long-read sequences that link the SSU, ITS and LSU turn the correspondence between VTs, SHs and LSU species from a guess into a measurement, and they show where each naming system misleads.

## 4.1 The three systems are nested rather than equivalent

SHs mostly fall within LSU species, which mostly fall within VTs. VT counts therefore underestimate the number of species (at best, VTs separate 19 of 37 species), and SH counts overestimate it, because the rRNA gene copies of one genome can fall into up to 6 SHs. Richness counted in SHs can thus be inflated by variation within genomes. How much it is inflated in a field

survey depends on which copies are amplified, detected, clustered and kept, which we did not test. Grouping the SHs of one genome restores most of the species signal in our cultures, but the groups were learned from these same cultures, and whether they transfer to field sequences is untested. The 2.5-kb amplicons were introduced to bring the three marker regions into one dataset [22]; what was missing is a measured correspondence between the names, which linked sequences supply.

### 4.2 The short-5.8S copies are genomic; whether they are functional is open

Variation among the rRNA gene copies of an AMF genome has long been known [41, 51], yet it has been largely overlooked in ecological and taxonomic work [46]. What linked sequences add is a measure of how this variation misleads each marker. Short-5.8S sequences look like new lineages in the LSU and form their own SHs and VTs, several of them under placeholder names in UNITE. The short-5.8S test recognizes most of them from their sequence alone, but less often when the whole species is absent from the reference, because no related culture with a 5.8S of the expected length is then available for comparison.

In *Rhizoglomus*, the short-5.8S sequences are not artifacts: the same 5.8S deletion recurs across isolates and species and in independent libraries of the same culture, and on trees the short-5.8S sequences of different species group together. The evidence is weaker for the other classes (Table 5). The short-5.8S sequences of *Septoglomus viscosum* are genomic but known from one species only, and the same-length sequences outside *Gigaspora* are single sequences of unresolved origin. A long deletion that recurs should therefore be read as part of a known genome rather than discarded as a singleton, whereas a divergent sequence seen only once remains an open case.

Being part of a genome does not make the short-5.8S copies harmless. A species that carries them has two structural variants of the marker, so the short-5.8S sequences form ITS clusters of their own (18 at 98.5%; Table S7). Moreover, the ITS and LSU suggest a different history from the SSU: on the LSU tree, the short-5.8S sequences of *Rhizoglomus intraradices* and *Rhizoglomus proliferum* group with *Sclerocystis* rather than with their own species. Because the class has a fixed signature, it can be flagged before clustering or tree building: in *Rhizoglomus*, a 13-nt motif in the 5.8S marks every short-5.8S sequence and no other. In UNITE, the same motif sorts the SHs of *Rhizoglomus* almost perfectly, and about one in five of these SHs consists only of the short type, so the paralog probably inflates SH counts in the public database as it does in the cultures. Such a flag has to be read together with the genus, because the motif is also the normal 5.8S of *Oehlia diaphana* and occurs in hundreds of unplaced Glomeraceae sequences, which only phylogenetic placement can sort out. The flag also needs the end of the 5.8S: ITS2 amplicons made with fITS7 or ITS3 include it, but an ITS2 trimmed with ITSx no longer does, so the flag applies to full-ITS and long-read sequences, and to ITS2 reads only before trimming. The motif was written from the sequences it was tested on, except for the UNITE search. Species are therefore best counted from main-type sequences, and the short type is best analysed as a locus of its own rather than mixed into a species tree.

Whether the short-5.8S copies are functional is a separate question. Their SSU keeps its base pairing, and their LSU, although rich in substitutions, mostly keeps it too. The deletion shortens one hairpin of the 5.8S, so under the rule we fixed in advance the class counts as structurally degraded, as pseudogenes would be. Yet the shortened stem consists of Watson–Crick pairs and is mostly closed by a UUCG loop, a combination known for stable hairpins [50]. The predicted folding thus shows a structural change, not a loss of function: DNA alone cannot distinguish a transcribed and processed copy from a silent one, and calling the class a pseudogene would go beyond the evidence. Sequencing the rRNA or its precursors from the same cultures would show whether the short type ends up in ribosomes. Same-length divergent sequences are more ambiguous: co-dominant types of one genome in *Gigaspora*, and single sequences of unknown origin elsewhere. They should be flagged rather than explained away.

### 4.3 No identity threshold delivers species

The distances at each level of relatedness show why the three systems disagree (Figure S1): their thresholds fall at different points on the scale of AMF variation. On the NS31–AM1 fragment, 97% identity would join almost every sequence with a congener (98.7%) and some with another genus, mostly in *Cetraspora*, *Gigaspora* and *Dentiscutata*, which fits the coarse resolution of VTs (at most 19 of 37 species). In the full ITS, the 98.5% threshold of SHs lies within the variation of a single culture, so one genome often spans several SHs. In the LSU, the 97% threshold of V18 matches the variation within species, but 54.5% of sequences have a congener within it. In no region does one threshold separate species, in line with the absence of a barcode gap that Stefani et al. reported for each region [46], so moving a threshold only trades lumping against splitting. Together, the regions do better: on the Krüger fragment, 97% identity splits few genomes and, outside the complexes, joins almost no species, and the whole sequence does the same at a higher identity. What no cut-off removes are the species complexes and the short-5.8S paralog. Combined names, with complexes reported as groups, avoid that trade-off where the markers allow it.

The SSU, by contrast, behaves predictably: between cultures, its distance was a median of 0.14 times the ITS distance, and no pair of cultures met the SSU-discordance test, so a sequence whose SSU differs by at least half as much as its ITS is unlike any pair of cultured relatives. The test caught reads with errors spread along them once the errors reached a few percent, but almost no chimeras, which keep an intact SSU from one parent. It therefore flags a possible artifact; it does not replace a chimera check, and an unusual but genuine lineage with no cultured relative could also trigger it.

### 4.4 Where the markers set a limit to novelty detection

SSU, ITS and LSU together cannot separate the members of 4 species complexes, 9 species in all: their closest sequences lie within each other's within-species limits on every marker. In the culture tests, all errors that remain after the ITS rule fall within these complexes at the 99th percentile used throughout. This is a limit observed in this reference, with complexes that the withheld species helped to define. As a prediction, the rule did well at the level of species and less well at the level of sequences: applied within each round of the LSU-removal test, it flagged every species that had errors and only 1 that had none, but 5 sequences of *Rhizoglomus irregulare* assigned to *Rhizoglomus vesiculiferum* lay outside that species' limit. In the LongGloDB test, whose predictions were fixed before the run, every false-known assignment was predicted; in the EUKARYOME test, 12 of 13 were not (see below). The same pair is the only exception at other percentiles: with the rule and the complexes both at the 95th or 97.5th percentile, *Rhizoglomus irregulare* and *Rhizoglomus vesiculiferum* are no longer a complex, and the sequences of *Rhizoglomus irregulare* assigned to *Rhizoglomus vesiculiferum* become errors outside a complex (Table S5). Whether these two names can be separated with rRNA genes is therefore the open question.

Within a complex, a species assignment is reliable only as a group, and novelty cannot be detected: an undescribed relative within the limit of a known species cannot be distinguished from that species with these markers. The alternative interpretation is that some of these names do not represent separate lineages, a question for taxonomists; *Diversispora aestuarii* and *Diversispora varaderana* differ by 0.0% in the SSU and 1.1% in the LSU. The rule depends on the data and will change as cultures are added: one further pair (*Diversispora celata* with *Diversispora eburnea*) misses the LSU limit by 0.4 percentage points.

The EUKARYOME test exposed two gaps in the rule. First, the rule compares a sequence with named species only. The 10 sequences labelled *Septoglomus constrictum* and *Septoglomus nigrum* lie inside reference cultures identified only to genus, and they took the names of the named species with which those cultures share a VT; the LSU clade of the unnamed cultures

could not object, because it is linked to no species. A reference culture without a species name is therefore not neutral: it lends its sequences to its named relatives. Such cultures should be identified, or a match to a name held only by unnamed cultures should block an assignment to species; we have not tested the second option. Second, LSU placement put the 2 sequences of *Dominikia achra* into the clade of *Dominikia duoreactiva*, although they lie outside its LSU limit. If the EUKARYOME labels are right, close relatives of reference species were misidentified at a higher rate (16.9% of sequences) than in the LSU-removal test (7.9%), although averaged over all absent taxa the rate was only 0.5%.

### 4.5 Novelty depends on what the reference already contains

With only the cultures withheld, LSU placement looks like a strong detector of new species, but much of that strength comes from the withheld species still being present in the LSU reference: its clade holds no culture, so its sequences are flagged as new. Phylogenetic placement was proposed because matching against a database cannot place sequences of taxa that are absent from it [11, 8]. This holds at the level of clades, but the LSU-removal test shows its limit at the level of species: with the species removed, a sequence is placed on its nearest named relative and then named after it, and with a genus removed, it is placed on the sister genus (69 *Rhizoglomus* sequences assigned to *Sclerocystis*). An undescribed species is absent from every reference. Removing it from the LSU reference is therefore closer to the real situation than leaving it in, but not equal to it: MaarjAM and UNITE were left intact, and many VTs are built from environmental sequences, so a species that is undescribed but already sequenced in the field can still have a VT of its own. The VT accounted for the novelty of 18 of the 673 sequences, so for a species never sequenced the false-known rate may be higher than reported, by up to about 2.7% of sequences. The ITS distance rule does not depend on external databases and closes part of the gap, but not where the ITS itself overlaps. Pooled over the test species, false-known assignments rose from 2.8% with the LSU reference intact to 11.6% with the species removed from it, and fell to 7.9% with the ITS rule, with wide intervals because 6 species accounted for all the errors. Each of these rates belongs to its reference condition, and none is the error rate for an undescribed species.

At the rank of genus, the same mechanism produces errors that look like answers. With a genus removed, its sequences were not classified as a new lineage but given the sister genus, and nothing in the assignment marked them. A genus that rests only on the nearest represented clade should therefore be treated with caution when the sequence lies far from every reference culture. A flag for such assignments, for instance based on the distance to the nearest culture of the assigned genus, is worth testing but was not part of the steps fixed here. The EUKARYOME test shows the scale of the problem: with their genus removed from the LSU reference, 16.4% of the sequences of absent genera were given a genus, against 3.7% with the genus present.

### 4.6 Relation to earlier work on the same cultures

Stefani et al. [46] sequenced the rRNA genes used here (148 cultures of 44 species) and compared them with three protein-coding genes (glomalin, $H^+$-ATPase and RPB1), sequenced for 143 cultures of 39 species. They found no barcode gap in any of the regions, so that no single distance threshold identifies AMF, and concluded that rRNA genes remain the practical choice because their primers work across all lineages. Our results agree where they overlap: the complexes are pairs whose closest sequences lie within each other's limits in every rRNA gene region. Two of their observations follow from the short-5.8S paralog. The rRNA genes of *Rhizoglomus vesiculiferum* and *Rhizoglomus proliferum* were paraphyletic in their trees, nested in *Rhizoglomus irregulare* and *Rhizoglomus intraradices*; on our concatenated tree, these four species become clades once the divergent sequences are set aside (Section 3.3). Moreover, their largest within-species distances on the Krüger fragment were in the species that carry the short type (*R.*

*irregulare* 7.7%, *R. vesiculiferum* 10.2%, *R. proliferum* and *R. intraradices* up to 11%) and in *Glomus chinense* and *Glomus rugosae* (7.4% and 8%), members of one of our complexes. Their distances are maximum pairwise distances over all sequences, with every gap counted, whereas ours are distances to the nearest sequence, so the numbers are not directly comparable. The questions also differ: they asked whether a distance threshold identifies a species within one marker, whereas we asked how the three naming systems relate and how far novelty can be detected once divergent copies are accounted for.

Their protein-coding sequences are public (glomalin PV808915–PV809055, RPB1 PV809056–PV809196, $H^+$-ATPase PV809197–PV809333) and could settle the status of the complexes: if those genes separate *Diversispora aestuarii* from *Diversispora varaderana*, the complex is a limit of the rRNA genes, and if not, it is a question of naming. We compared them (Table S6). At the thresholds Stefani et al. proposed, the three genes separate almost every pair of congeners that the rRNA genes separate (42 of 44 on glomalin; 43 of 45 on RPB1; 43 of 45 on $H^+$-ATPase; 95–96%). Of the 6 pairs of species within complexes, 4 have protein-coding records for both species. Of these, 2 exceed the threshold on one gene each (*Diversispora aestuarii* and *Diversispora varaderana* on RPB1 (1.24% against 1.1%); *Rhizoglomus irregulare* and *Rhizoglomus vesiculiferum* on glomalin (1.32% against 1.0%)), but none of the 12 comparisons exceeds the variation within the species themselves. For the pairs with records, the complexes are therefore not a limit of the rRNA genes alone: on these genes, too, their members lie as close as cultures of one species, and whether they are separate species is a question for taxonomy rather than for the choice of marker. The pairs without records, and other genes or criteria, could still separate them. Linking the records through culture names also showed that the RPB1 and $H^+$-ATPase deposits exchange the species names of three cultures: FS16_2427A (*Diversispora epigaea* in the reference, *Septoglomus* sp. in GenBank); FS_9_2328B (*Septoglomus* sp. in the reference, *Rhizoglomus vesiculiferum* in GenBank); Gwalkeri_IVT132A (*Rhizoglomus vesiculiferum* in the reference, *Diversispora epigaea* in GenBank). Each of these records lies 0.0–0.9% from records of its culture's species, so the sequences are correct and the names are not; we left the 6 records out.

### 4.7 Why the rules are not calibrated per species

A calibrated rule for deciding when an ITS sequence is new exists for fungi in general [37]. Here, the limit of the ITS distance rule is a quantile of the reference, the 99th percentile of the distance between cultures of one species, and within the reference it behaved as intended: it changed 6 of the 755 sequences of known species to 'new' (0.8%). It is nonetheless an empirically chosen decision threshold, not a calibrated error rate: in cultures from another collection, it changed 9.1% of the ASVs of known species and prevented no error (Section 3.8). The 99th percentile was fixed before the test species were run. The 95th percentile would have reduced false-known assignments further (5.5% against 7.9% with the species removed from the LSU reference) at the cost of changing 4.2% of the sequences of known species instead of 0.8% (Table S5), a trade-off that users can set, and should set with their collection in mind. The other rules, such as the change to 'new' when a VT or LSU name is unmatched, are not calibrated, because the reference holds too few cultures per species to fit a threshold for each, and because the problems peculiar to AMF, divergent copies and species complexes, have to be handled before a calibration can mean anything.

### 4.8 Independent tests

On 52 cultures from three other laboratories, with the predictions fixed before the run, the rules assigned a species absent from the reference to a known species only where the complex rule had predicted it: *Gigaspora gigantea* KS302, which lies inside *Gigaspora rosea* on every marker. The test is easier than ours in one respect and harder in two. Its absent species are distant from those

of the reference, so it says little about the sister-species case, which occurred once. Its amplicons start inside the VT fragment, which leaves the VT search silent unless coverage is counted over the part that a sequence spans, and a genus taken from the VT alone was then wrong when the true genus was absent from the reference (*Racocetra* given *Cetraspora*). Finally, its cultures come from other collections, some of which lie farther from ours in the ITS than our cultures of one species lie from each other: the ITS distance rule, fitted at the 99th percentile on one collection, changed 15 of the 164 ASVs of known species to 'new' and prevented no false-known assignment.

The EUKARYOME test complements it with field and reference sequences of many more taxa, including close relatives of reference species, but its names are a classification rather than culture names. Here the ITS distance rule did prevent errors (11), at the cost of 20 sequences of reference species that it changed to 'new', all of which lay outside their labelled species on at least one marker. Whether those sequences are mislabelled or show real variation within species cannot be decided from EUKARYOME. In either case, the limit of the rule should be fitted on more than one collection before the rule is used on field data.

### 4.9 Implications for surveys

We recommend naming taxa at the level that the markers support: report VTs, SHs grouped by genome and LSU species clades, with species groups where the markers cannot decide, and do not expect any identity threshold to deliver species on its own. Screen for the short-5.8S type before counting taxa or building trees, for instance with the motif in *Rhizoglomus*: in EUKARYOME, the decision steps recognized only 46 of the 95 sequences of *Rhizoglomus* that carry it, and classified 28 as new lineages. Treat paralog SHs and VTs, such as SH1337353 and VTX00409, as parts of known genomes. Identify reference cultures to species, or keep cultures without a species name from lending species names to the sequences that match them. Base claims of novelty on evidence that does not come from the database that defines what is known, report error rates together with the reference they were measured against, and interpret names according to a stated taxonomy [48], because outdated names in a database look like errors of the marker [7]. Treat the SSU-discordance flag as a check for noisy reads rather than for chimeras, and a genus given to a sequence that lies far from every reference culture as provisional. Replication across independent sequencing libraries is still needed for environmental data.

### 4.10 Limitations

Apart from the LongGloDB and EUKARYOME tests, all results come from one collection of cultured species, and these cultures were used to develop the methods, so the results are development estimates. The LongGloDB test rests on 26 cultures of absent species, covers only distant novelty, and used amplicons on which the VT search, as specified, is silent. The EUKARYOME test covers close relatives, but only 77 of its sequences belong to absent species in reference genera; its names are a classification, less reliable at species level than culture names, so some of its errors may be errors of the labels; and EUKARYOME is curated by one of us. The 99th percentile behind the ITS rule and the complexes was chosen after the distances in these cultures had been seen (Table S5 shows the effect of other choices), and the complexes were derived from all cultures, including the withheld species; applied within each round, the rule anticipated the species at risk but not every misidentified sequence. Sequences, cultures and species are nested, and intervals that resample species are wide (Table 12). Culture names are treated as correct, although some are doubtful: *Microdominikia litorea* 5274, some *Gigaspora rosea* cultures, and *Septoglomus* sp. C4 and 5070_5024A, whose LSU matches *Microviscospora peruviscosa*. One of the 6 sequences of known species changed to 'new' by the ITS rule comes from *Microdominikia litorea* 5274, and the reference's *Ambispora leptoticha*, ON205A, is *Ambispora gerdemannii* in INVAM, so some apparent errors may be errors of the names, and agreement

with a deposited name does not prove the species. Recurrence across culture labels can overstate independence, because one isolate may be held by several collections under different codes, and only 16 of the 51 divergent sequences come from cultures with replicate libraries. MaarjAM and UNITE stayed intact in every test, and the LSU-removal test removed species from the LSU reference only. About a quarter of the V18 species names with several records (51 in the placement audit) do not form exclusive clades on its backbone tree. The structure analysis reads DNA: a copy that keeps its base pairing need not be transcribed or end up in ribosomes, and a copy that loses a pair need not be silent. The distances at each level of relatedness are exploratory and fix no rule. Replication on environmental sequencing libraries is outstanding.

## 5 Conclusions

Long-read rRNA gene sequences that link the SSU, ITS and LSU provide an empirical bridge between the three AMF naming systems and show where each one misleads. Detecting novelty is only as reliable as the reference databases are complete, and it cannot go below the resolution of the markers: with the withheld species removed from the LSU reference, 7.9% of their sequences were still assigned to a known species, all of them within species pairs that SSU, ITS and LSU cannot separate. As a prediction, the complex rule flagged the species at risk, and in cultures from other collections the only species misidentified as known was the one it predicted, but it did not anticipate every misidentified sequence. In EUKARYOME, it missed most errors, which arose through reference cultures identified only to genus. Divergent copies must be recognized before taxa are counted: in *Rhizoglomus*, the short-5.8S type is a paralog family shared across species, whereas the origin of other divergent sequences is less certain. Species that the markers cannot separate, and that the available protein-coding sequences do not separate either, should be reported as species complexes.

## Data and code availability

The culture sequences are GenBank PX207096–PX207274 and PX214440–PX215164. The code and data behind every result are at `https://github.com/ayobi/amf-novelty-benchmark`, archived as release 0.3.1 at doi:10.5281/zenodo.23422616 [36]. The release holds the culture reference (version 4, `181938d695f3df40`), the curation edits and their manifests, the scripts, the outputs of every test, the protocols with their checksums, the fixed predictions, and the checks of divergent sequences with their alignments and trees (Supplementary Data S1). The analyses added after review are `stage3/artifact_control.py` and `paper/analysis/foldwise_complexes.py`, `intervals.py` and `divergent_classes.py`. Earlier releases archive the state of the repository with the LongGloDB test package (release 0.1.0, doi:10.5281/zenodo.23219691 [34]) and with the checks of divergent sequences (release 0.2.0, doi:10.5281/zenodo.23374725 [35]). No LongGloDB sequence is redistributed; LongGloDB v1 is available from GloDB [14]. The EUKARYOME test stores sequence identifiers only; EUKARYOME v2.1 is available from its authors under CC BY 4.0 [47]. External references: MaarjAM gDAT snapshot (sha256 `2b2c57e664e76347...`), UNITE 10.0 of 19 February 2025 [1] (UNITE+INSD sha256 `abe98f43e16c179f...`), Delavaux V18 AMF-only (sha256 `e6794cbbe166735a...`), and the Rfam models RF01960 and RF02543 (downloaded on 12 October 2026; sha256 `2cd46b2a3fe3f463...` and `63b122094f59d229...`) and RF00002 (sha256 `bc55434ebc3ef8c3...`), all three built in July 2024.

## Author contributions

Aaron O'Brien conceived the study, developed the software, carried out the analyses and wrote the manuscript. César Marín reviewed the manuscript, in particular its treatment of AMF

taxonomy and biology. Leho Tedersoo reviewed and edited the manuscript, in particular its terminology, nomenclature and clarity for mycologists, and suggested further analyses. All authors approved the final version.

## Acknowledgements

This study rests on the long-read rRNA gene sequences of identified cultures that Franck Stefani and his co-authors generated and made public [46]; we thank them for that work, and Franck Stefani for welcoming its use here. We also thank the maintainers of UNITE, MaarjAM, EUKARYOME, Rfam and GloDB (the Delavaux V18 and LongGloDB databases).

## Competing interests

Aaron O'Brien develops ITSxRust [33], which is used in this study. The authors declare no other competing interests.

## Use of generative AI

Anthropic's Claude and Microsoft Copilot (the versions current in 2026) were used in preparing this work: to draft and revise manuscript text and to write and debug analysis scripts. Aaron O'Brien revised the drafted text extensively, ran all analyses, and verified all reported values against the primary output; the authors are responsible for the content of the manuscript.

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# Supplementary material

## Methods S1. Marker fragments

ITSxRust anchored each read with one nhmmer pass (HMMER 3 [13], ITSx fungal profile [3]) and delimited the SSU region, ITS1, 5.8S, ITS2, full ITS and LSU region. We used tolerant anchoring thresholds (E-value $\leq 10^{-3}$, anchor score $\geq 15$), because anchoring only delimits regions. A panel of seven fragments (Table S1) was cut at the original primer sites, so that each fragment covers the region on which its database was built. The panel used the primers of the SSU virtual taxa (NS31, AM1 and AML2 [45, 17, 27, 40]), the SSU V4 primers WANDA and AMV4.5NF/AMDGR [12, 42], the LSU primers LR0R [5, 11] and FLR2 [49], the D2 primers FLR3 and FLR4 [15], and the outer pair AML1 and LSUmBr [27, 26] of the long amplicon [22], with LSUmBr as the degenerate consensus of its five variants [43]. Primers were located with an IUPAC-aware infix alignment (edlib [54]) that tolerated edits up to 15% of the primer length; fragments were primer-trimmed, and coordinates are 1-based, inclusive and in rRNA-sense orientation. Every primer sequence was checked against published sources (`analysis/primer_check.tsv`). A primer was searched for only in the part of the read where it can lie (SSU primers before the ITS, LSU primers after it, plus 50 nt). A fragment is *open* when a primer site lies beyond the end of the sequence, and *missing* when the primer is absent although the sequence covers its expected position. Sequences were also screened for repeats (concatemers) with the concatemer screen of version 0.3.1 of our software. No fragment crosses an ITSxRust anchor, and no sequence was flagged as a repeat.

## Methods S2. Culture reference

The names of the deposit are those of the culture tables of Stefani et al. (their Tables S2 and S3). Their 39 protein-coding species are phylogenetic species, which split *Septoglomus* sp. into three; their 44 rRNA gene species are not itemized, and we used the names of the deposit, with the species deposited in *Rhizophagus* given in *Rhizoglomus* [44]. A culture is an organism name plus a culture label. The reference was built by `freeze_ref.py` from the fragment tables and a file of edits. Every edit first stated what the data had to look like (organism, label, number of sequences, VT, sequence class, 5.8S length), and a failed check stopped the build before anything was written. Each version recorded the checksums of its input and output files and a fingerprint. The edits leading to version 4 (fingerprint `181938d695f3df40`) are listed in Table S2. The sequence classes follow the 5.8S: sequences whose 5.8S lies more than 3% from that of the culture's medoid were flagged, and the flagged sequences were split into short-5.8S sequences, at least 3 bp shorter than the culture's main type, and same-length sequences. Where the medoid fell inside the short-5.8S group, the flags came out inverted, and the classes were swapped (Table S2).

## Methods S3. Searches and near-best names

The VT search used the MaarjAM gDAT snapshot of July 2020 [40], the SH search UNITE release 10.0 of 19 February 2025 [32, 1], and the LSU search the Delavaux V18 database of May 2025 [11, 8, 9] (repository commit `f4a0014`). Searches used BLAST+ [6] with the settings in Table 1. Many V18 records carry terminal N padding, so coverage did not count reference Ns; without that correction, 95 of the 460 records would fail the coverage test even against themselves. Hits to any accession of the culture reference were disregarded. The name of a culture in a system is the most common name of its main-type sequences (ties went to the higher median identity). The names of a sequence depend only on its fragment, the database and the disregarded accessions, not on culture names. The comparison of the names (the crosswalk) was run before the last two *Septoglomus* edits (Table S2), so its report on cultures was replayed on the final labels with the stored names of each sequence (`crosswalk_replay.py`), after every

stored name had been checked against evidence captured on the final reference: in all three systems and for all 903 sequences, the stored identity equaled that of the highest-scoring hit passing the threshold, and the best name was present at that identity.

Each sequence received a set of near-best names instead of a single best hit. The anchor was the sequence's best hit by bit score, and the set held every name within $\delta$ identity points of the anchor that also passed the identity threshold. Names that split the main-type sequences of one culture at $\delta = 0$ were merged when the closest sequences across the split were within 1% in the SSU region; this limit prevents the two lineages of a mixed culture from being merged. $\delta$ was set to the smallest margin at which the main-type sequences of at least 99% of cultures share a name; for VTs, it was also at least the 95th percentile of the SSU differences within cultures. SHs and LSU names have no such floor, because their variation within genomes is wide and uneven (95th percentile: ITS 8.6%, LR0R–FLR2 3.9%), and the merged names absorb that variation instead (Table S3).

## Methods S4. LSU placement

The V18 reference sequences were realigned with MAFFT [21], and branch lengths and GTR+$\Gamma_4$ parameters were fitted on the published topology with RAxML-NG [24]. Query sequences were added with `-addfragments -keeplength` and placed with EPA-ng [2]. The compatible edges of a query carry at least 95% of its likelihood weight. Edges inside a maximal clade bearing a single V18 species name were collapsed into one name (the LSU species clade). Where a species name is not monophyletic on the backbone tree, its separate components stayed separate names, and edges that carry several species names stayed individual names.

## Methods S5. Combining the names and the decision steps

A name was linked to the species and genera of the training cultures whose sets of near-best names hold it. In each system, a sequence is *matched* when its names belong to training cultures, *unmatched* when it has a name but no training culture holds it, and *silent* when it has no name. Genera were intersected over the matched systems, and species over the matched systems whose names are linked to named species. The repository's codes for the decision steps are: SSU discordance (possible artifact), A1; divergent sequence, D1 (short 5.8S) and D2 (paralog name), with D3 for a same-length 5.8S that is only flagged; near a VT threshold, B1; known species or species group, K1 (all matched systems agree and neither the VT nor the LSU name is unmatched); new species in a known genus, N1 (VT unmatched), N2 (LSU unmatched), N3 (no species remains while a genus does) and N6 (the ITS distance rule); discordant, X1 (the matched systems share no genus); and new lineage, N4 (no system matched) and N5 (as N4, with the best VT also below the near-threshold band). The comparators of D1 are the main-type sequences of the nearest training culture; the support of D2 is the number of other training cultures that reach the name only through short-5.8S sequences. Divergent sequences are named by the VT alone. A sequence is scored against what the reference contains: main-type sequences of species held by another training culture should receive a known species or species group that contains their species; main-type sequences of withheld species should not receive a known species (otherwise a false-known assignment); and divergent sequences should be recognized as divergent or, for same-length sequences, at least not be considered new.

## Methods S6. Withholding tests and the choice of percentile

Every quantity derived from cultures was fitted on the training cultures of each round only: merged names, $\delta$, the links from names to species, the comparators of D1, the support of D2, the pools of nearest sequences and the limit of the ITS distance rule. The test with withheld cultures had 148 rounds, each removing one physical culture label together with all lineages

sharing it; the test with withheld species had 50 rounds, one per named species with all its cultures removed plus one per culture not named to species. In these tests, MaarjAM, UNITE and V18 stayed fixed. In the LSU-removal test, V18 was realigned without the target's records, the model refitted and all sequences placed again (Table S4). The second version of the protocol added the ITS distance rule (N6) and changed nothing else; *Glomus chinense* and *Glomus rugosae* were used to develop that rule and are reported as development rounds. The test species (the 'first-outcome' rounds in the repository) are all other species: the 34 whose V18 records were removed, and *Ambispora callosa*, which V18 does not hold by name, reported apart. The same protocol had 15 genus rounds, in which all cultures of a genus and the genus's V18 records were withheld together.

*The complex rule within each round* (added after review; `analysis/foldwise_complexes.py`). For each species round of the LSU-removal test, the cultures withheld in that round (`holdout_plan.json`) were removed, the within-species limit of each marker was recomputed from the remaining main-type sequences of named species as in `species_groups.py`, and each main-type sequence of the withheld species was compared with the deduplicated main-type sequences of every remaining named species. A sequence lay inside a species when its distance to the nearest sequence of that species was at most the limit on all three markers, the rule used for the locked predictions. With nothing withheld, the recomputed limits equal those of the main text. The withheld sequences served only as queries. False-known assignments per species were taken from Table S4, and matched to individual sequences with the per-sequence results of the run.

*Intervals* (`analysis/intervals.py`). The withheld-culture and withheld-species tests were rescored from the frozen tables of the decision steps with the frozen ITS distance rule, refitted per round, and every count was checked against the values reported here. For each rate, species (or cultures) were resampled with replacement 10,000 times (seed 20261015) and the rate pooled over the resampled units; the intervals are the 2.5th and 97.5th percentiles. For the LSU-removal test, intervals over species come from the per-species counts and intervals over cultures from the per-sequence results, with a random stream of their own, so that adding them changed no other interval.

To test the choice of percentile, the ITS distance rule and the complexes were set together at the 95th and 97.5th percentiles (Table S5): the rule was re-applied to the stored results of the decision steps, with the limit of each round refitted on its training cultures (`percentile_sensitivity.py`). Percentiles follow Python's exclusive quantile method throughout, as in the complex rule (`species_groups.py`).

## Methods S7. The independent tests

LongGloDB v1 holds 1,122 ASVs. We left out the 810 ASVs that carry the laboratory codes of the Stefani et al. cultures (STFP and STFI): 784 of them are identical in full ITS to a sequence in our reference, so they are the same cultures processed again. A further 4 cultures share a culture code with a Stefani label (3 of them under another species name) and were left out too. This overlap was established with `longglodb_overlap.py`. The protocol and its two amendments (`PROTOCOL_LGDB.md`) fixed the outcomes, conditions and analyses, and the predictions of the complex rule were fixed by their checksums before any test sequence was searched, placed or scored. The frozen code of the second protocol version was used unchanged. All 151 reference cultures were used for training and every test culture was excluded, so no test sequence entered merges, margins, name links, comparators or the limit of the ITS distance rule. Test sequences were cut by the same fragment step as the culture sequences, and their VT and SH evidence was computed with the same filtering against the same releases. In condition A, the test fragments were placed on the intact V18; in condition B, each of the 19 absent species, all of which V18 holds by name, was removed from V18 in its own round, as in the LSU-removal test. The second analysis (amendment 2) measured VT coverage over the part of each MaarjAM reference that

the query spans.

The EUKARYOME test (`PROTOCOL_EUK.md` and its note 1, in `benchmarks/eukaryome`) ran the same frozen code, through a wrapper that replaces only the reader of the test set, the plan of condition B and the culture identifiers, with the same parity checks. The name table, roles and removals were made by `eukaryome_overlap.py`, and the sequences were cut by the fragment step with the options used for the culture sequences. The predictions were computed by the same script as for LongGloDB, with the within-species limits passed at the definition of the main text (ITS 5.99%, LSU 3.03%, SSU region 0.81%), so the two definitions noted for LongGloDB do not arise. Condition B removed 51 sets of V18 records, one per absent species or genus that V18 holds, with every record of a genus removed in its round. The run outputs, without sequences, are in `benchmarks/m4_external_holdout/runs_euk`.

## Methods S8. Trees, artifact check, rRNA structure and distances

The protocol (`benchmarks/copy_checks/PROTOCOL_COPY_CHECKS.md`) was fixed on 11 October 2026 and the structure analysis (`ADDENDUM1_FUNCTION.md`) on 12 October 2026, each before its results were computed.

*Phylogenetic trees.* The sequences of each order (Archaeosporales 30, Paraglomerales 28, Diversisporales 245 and Glomerales 600) were analysed separately, because the ITS cannot be aligned across orders and every culture and species lies within one order. MAFFT v7.505 L-INS-i (`-localpair -maxiterate 1000`) was used without trimming; the concatenated alignment joins the three alignments with one partition per region. IQ-TREE 2.3.6 was run with ModelFinder (`-m MFP`; for the concatenated alignment, one model per partition with edge-linked proportional branch lengths), 1,000 ultrafast bootstraps (`-B 1000`), 1,000 SH-aLRT replicates (`-alrt 1000`), near-zero branches collapsed (`-czb`) and seed 20261011. Each tip is one distinct sequence and stands for every identical sequence. A group is tested on a tree when it has at least two sequences. It is *unresolved* when no branch separates it and none contradicts it (it sits in a polytomy, or shares an identical sequence with another group); a branch that only joins identical sequences has no support value and never counts as strong. For a group that is not monophyletic, its *escapees* are its sequences outside the largest clade made only of its sequences. All 1,436 tests of monophyly were recomputed independently with Bio.Phylo, with the same result.

*Artifact check.* Each sequence was aligned to the nearest main-type sequence of its own culture (distance over the whole sequence; never itself) by global alignment with free end gaps (Biopython `PairwiseAligner`; match 2, mismatch −3, gap opening −5 and extension −2, the BLASTN defaults). Gaps that touch either end of the alignment reflect differences in read length and were ignored. Each internal deletion was assigned to the region of the comparator (SSU region, ITS1, 5.8S, ITS2, LSU region) that holds most of its bases. Two sequences are of the same type when their full-ITS distance is at most 1.0%, and the two lineages of a mixed label count as one isolate. Sensitivity analyses fixed in advance: same type at 0.5% and 2.0%, and long deletions read as at least 10 nt in every region.

*Controls for the SSU-discordance flag* (added after review; `stage3/artifact_control.py`, results in `benchmarks/copy_checks/artifact_control`). The first decision step was re-implemented exactly as frozen: the nearest main-type sequences of other cultures in the SSU region, with ties, and the flag when the SSU distance is at least 1% and at least half the full-ITS distance to every tied sequence. Queries: every main-type sequence, with its culture label withheld; one main-type sequence per culture label, with random errors at 0.5, 1, 2, 3, 5 and 8% of positions (60% substitutions, 20% insertions, 20% deletions) applied to the SSU region and the full ITS alike; and chimeras of main-type sequences of two different named species, 150 with parents in different genera and 150 with congeneric parents of each kind, either joining the SSU region of one to the full ITS of the other or switching parents at a random point in the middle 80% of the SSU region. Each query was scored with the culture labels of its parents withheld. Seed 20261015; 95% intervals are Wilson intervals.

*rRNA structure.* Infernal 1.1.5 `cmalign` was run with truncated alignments allowed. The groups were the short-5.8S sequences, the same-length sequences, the main-type sequences with a long deletion in the SSU region, 5.8S or LSU region (*flagged main*), and all other main-type sequences with a main-type culture-mate (the baseline, which set the 99th percentiles). The paired comparison set each divergent sequence against the mean of the main-type sequences of its culture (two-sided Wilcoxon signed-rank test). The 5.8S deletions of the short-5.8S sequences were located on the RF00002 consensus. The ITS2 was folded with ViennaRNA 2.7.2 `RNAfold` (default parameters, 37 °C): minimum free energy per nucleotide, and the base-pair distance between the structure of a sequence and that of its comparator.

*Distances.* For pairs of cultures, each culture was represented by its medoid main-type sequence, the sequence with the smallest summed full-ITS distance to the other main-type sequences of its culture. The within-species limit of the NS31–AM1 fragment (1.7%) was computed as for the three markers of the complex rule. The 28 Paraglomerales sequences lack the NS31–AM1 fragment and were left out of that region.

*Longer regions.* The Krüger fragment was cut from each sequence from the SSUmAf1 site [26] (found with at most one mismatch in all 903 sequences) to the end of the sequence at LSUmBr, so the last stretch of the SSUmAf–LSUmAr amplicon is missing; the whole sequence runs from AML1 to LSUmBr. Distances, classes and limits are as above. For a cut-off $t$, a main-type sequence is split from its culture if it lies more than $t$ from every other sequence of its culture, and joined to another species if a sequence of another species of its genus lies within $t$; the partners of the four complexes are counted both ways (Figure 4). Script: `stage3/copy_long_fragments.py`.

*What the short type does.* Short-5.8S sequences were counted by their 5.8S sequence: the short-5.8S class and every sequence whose 5.8S is identical to one of them (this adds PX214776.1). All retained sequences of the named species were clustered by single linkage within each order at 98.5% and 97% identity on the full ITS, the Krüger fragment and the whole sequence, once with every sequence and once with main-type sequences only (Table S7). On the Glomerales tree of each marker, rooted on a *Funneliformis* tip, we recorded the clades made only of short-5.8S sequences, their support, the taxa they join and the main-type taxa of their sister group. Script: `stage3/copy_paralog_effects.py`.

*A motif for the short type.* The motif YTYGGRGARTATG covers the short-type hairpin of every short-5.8S sequence of *Rhizoglomus*. It was written from these sequences and tested on the same sequences, so it shows that such a flag is possible, not its error rate. We searched for it in the 5.8S and in the whole sequence of all 903 retained sequences, and located the ITS2 primers fITS7 [19] and ITS3 [53] in the 5.8S, allowing up to two mismatches (ITS3 matched every *Rhizoglomus* sequence with one or two). Script: `stage3/short58_motif.py`; results in `benchmarks/copy_checks/short58_motif`. As an independent test, we searched for the motif in all 2,069,189 sequences of the UNITE+INSD set of 19 February 2025 [1], which holds none of the culture sequences. A hit counted as in the 5.8S when fITS7 (up to two mismatches) started 30–45 nt before it, as in the culture sequences (37 nt), and a sequence spanned the site when fITS7 was found with at least 60 nt after its start. SHs are the SH codes of the UNITE records. Script: `stage3/short58_motif_db.py`; results in `benchmarks/copy_checks/short58_motif/unite`. These analyses are exploratory and fix no rule.

Table S1: Marker fragments cut from the 904 deposited rRNA gene sequences (primer-trimmed fragments; primers found with up to 15% of their length in mismatches or gaps; software version 0.3.1).

| Fragment | Used by | Usable | Missing | Open | Median (bp) |
|---|---|---|---|---|---|
| SSU NS31–AM1 | MaarjAM VTs | 876 | 28 (3′) | 0 | 508 |
| SSU NS31–AML2 | SSU studies | 904 | 0 | 0 | 522 |
| SSU WANDA–AML2 | SSU V4 | 903 | 1 (5′) | 0 | 500 |
| SSU AMV4.5NF–AMDGR | SSU V4, short | 904 | 0 | 0 | 216 |
| LSU LR0R–FLR2 | LSU reference V18 | 904 | 0 | 0 | 735 |
| LSU FLR3–FLR4 (D2) | LSU D2 | 904 | 0 | 26 (3′) | 330 |
| AML1–LSUmBr | full amplicon | 0 | 904 (both) | 0 | – |

Usable: complete plus open-ended fragments; missing: primer absent although the sequence covers its position; open: primer site beyond the end of the sequence. The 28 missing NS31–AM1 fragments are all the Paraglomerales sequences, in which AM1 does not bind; the one missing WANDA fragment is the sequence dropped from the reference (PX215117.1). The 26 open D2 fragments are sequences that end before the FLR4 site. The outer amplicon primers are absent because the deposited sequences are trimmed inside them. No fragment crosses an ITSxRust anchor, and none of the 904 sequences is flagged as a concatemer.

Table S2: Curation edits made to the deposited sequences to build version 4 of the culture reference.

| Item | Edit | Evidence |
|---|---|---|
| PX215117.1 (*Gigaspora rosea*) | Dropped | SSU and ITS diverge at the same rate (4–6% and 5.6–7.6%): error or pseudogene |
| BR105_2435A | Second lineage split off as *Paraglomus* sp. | Three VTX00281 sequences 28–33% from the culture's *Paraglomus brasilianum* sequences in ITS |
| 4375_IVT88A, ON205A | Labels merged | One culture filed under two organism names |
| T3_IVT34A, CC4_4A_IVT7A | Kept as *Rhizoglomus irregulare* | ITS2 a median 3.3% from *Rhizoglomus irregulare* and 6.4% from *Rhizoglomus vesiculiferum* (T3); nearest LSU reference *Rhizoglomus irregulare* |
| SD5_IVT32, 212939_IVT38A | Main and divergent classes swapped | Medoid fell in the short-5.8S group |
| SO43BL, Blasz_17_2277A | Mixed cultures split; SO43BL named *Septoglomus africanum* | Reciprocal lineages; ITS and LSU agree on *Septoglomus africanum* |

Table S3: Near-best names of the culture sequences in the three systems (culture reference version 4).

| System | Merged names | Margin $\delta$ | Consistent cultures | One name | Species resolved |
|---|---|---|---|---|---|
| VT | 2 | 0.6 | 138/138 | 74% | 16 of 37 (19 at $\delta = 0$) |
| SH | 35 | 0 | 106/107 | 100% | 32 of 37 |
| LSU (nearest reference) | 5 | 0 | 128/128 | 81% | 27 of 37 |

A culture is consistent when the near-best names of its main-type sequences share a name. *One name*: share of main-type sequences with a single near-best name. A named species is resolved when no other named species shares any of its names. Merged names and species counts are learned and scored on the same cultures, so they are upper bounds.

Table S4: False-known assignments for each withheld species in the LSU-removal test.

| Withheld species | Sequences | LSU ref. intact | | species removed | |
|---|---|---|---|---|---|
| | | no rule | ITS rule | no rule | ITS rule |
| *Acaulospora morrowiae* | 10 | 0 | 0 | 0 | 0 |
| *Ambispora callosa*‡ | 20 | 20 | 0 | 20 | 0 |
| *Ambispora leptoticha* | 10 | 10 | 0 | 10 | 0 |
| *Cetraspora gilmorei* | 60 | 0 | 0 | 0 | 0 |
| *Dentiscutata heterogama* | 6 | 0 | 0 | 0 | 0 |
| *Diversispora aestuarii* | 20 | 1 | 1 | 20 | 20 |
| *Diversispora arenaria* | 4 | 0 | 0 | 4 | 4 |
| *Diversispora celata* | 10 | 0 | 0 | 0 | 0 |
| *Diversispora eburnea* | 15 | 0 | 0 | 0 | 0 |
| *Diversispora epigaea* | 13 | 0 | 0 | 0 | 0 |
| *Diversispora jakucsiae* | 4 | 0 | 0 | 4 | 4 |
| *Diversispora spurca* | 18 | 0 | 0 | 0 | 0 |
| *Diversispora varaderana* | 31 | 8 | 8 | 3 | 3 |
| *Dominikia bonfanteae* | 6 | 0 | 0 | 0 | 0 |
| *Dominikia duoreactiva* | 9 | 0 | 0 | 0 | 0 |
| *Funneliformis caledonium* | 9 | 0 | 0 | 0 | 0 |
| *Funneliformis coronatus* | 24 | 0 | 0 | 0 | 0 |
| *Funneliformis geosporum* | 8 | 0 | 0 | 8 | 0 |
| *Funneliformis mosseae* | 130 | 0 | 0 | 0 | 0 |
| *Gigaspora rosea* | 51 | 0 | 0 | 0 | 0 |
| *Glomus chinense*† | 35 | 28 | 5 | 28 | 5 |
| *Glomus macrocarpum* | 7 | 0 | 0 | 4 | 4 |
| *Glomus rugosae*† | 75 | 4 | 4 | 75 | 62 |
| *Glomus tetrastratosum* | 21 | 0 | 0 | 0 | 0 |
| *Microdominikia litorea* | 12 | 0 | 0 | 0 | 0 |
| *Oehlia diaphana* | 29 | 0 | 0 | 0 | 0 |
| *Paraglomus brasilianum* | 8 | 0 | 0 | 0 | 0 |
| *Paraglomus laccatum* | 5 | 0 | 0 | 0 | 0 |
| *Rhizoglomus intraradices* | 12 | 0 | 0 | 0 | 0 |
| *Rhizoglomus irregulare* | 56 | 0 | 0 | 19 | 18 |
| *Rhizoglomus proliferum* | 6 | 0 | 0 | 0 | 0 |
| *Rhizoglomus silesianum* | 7 | 0 | 0 | 0 | 0 |
| *Sclerocystis sinuosa* | 18 | 0 | 0 | 0 | 0 |
| *Septoglomus africanum* | 6 | 0 | 0 | 6 | 0 |
| *Septoglomus turnauae* | 4 | 0 | 0 | 0 | 0 |
| *Septoglomus viscosum* | 21 | 0 | 0 | 0 | 0 |
| *Rhizoglomus vesiculiferum* | 23 | 0 | 0 | 0 | 0 |

Main-type sequences of each withheld species that were assigned to a known species, with the LSU reference (V18) intact or with the species removed from it, each without and with the ITS distance rule. † development species; ‡ species not held by V18 under its name (removal changes nothing).

Table S5: The ITS distance rule and the species complexes recomputed together at three percentiles.

| Percentile | ITS limit (median) | Complexes (species) | Known demoted (of 755) | False known species withheld (of 803) | False known also LSU-removed (of 673) | Outside a complex |
|---|---|---|---|---|---|---|
| 95th | 4.4% | 3 (6) | 32 (4.2%) | 9 (1.1%) | 37 (5.5%) | 5 |
| 97.5th | 5.0% | 3 (7) | 22 (2.9%) | 13 (1.6%) | 53 (7.9%) | 18 |
| 99th | 6.0% | 4 (9) | 6 (0.8%) | 18 (2.2%) | 53 (7.9%) | 0 |

The ITS distance rule is re-applied to the stored results of the decision steps at each percentile, with the limit fitted on the remaining cultures of each round, and the complexes are those of the complex rule at the same percentile. *Known demoted*: sequences of known species that the rule demotes with one culture withheld. *Also LSU-removed*: the test species of the LSU-removal test. The 99th-percentile row reproduces the reported results exactly. Every error outside a complex is *Rhizoglomus irregulare* assigned to *Rhizoglomus vesiculiferum*.

Table S6: Distances in protein-coding genes between the members of the species complexes: smallest distance (%) between the two species, with the largest distance within either species in brackets.

| Pair | Glomalin | RPB1 | $H^+$-ATPase |
|---|---|---|---|
| *Diversispora aestuarii* / *Diversispora varaderana* | 0.73 (0.0) | 1.24 (1.73) | 1.22 (0.0) |
| *Diversispora arenaria* / *Diversispora jakucsiae* | 0.29 (–) | 0.57 (–) | 1.55 (–) |
| *Glomus chinense* / *Glomus rugosae* | 0.73 (0.73) | 0.33 (7.93) | 1.00 (1.32) |
| *Rhizoglomus irregulare* / *Rhizoglomus vesiculiferum* | 1.32 (1.61) | 0.81 (7.54) | 1.18 (5.77) |

Sequences of Stefani et al. (2025). Indicative species thresholds from that study: 1.0% / 1.1% / 1.7% (glomalin / RPB1 / $H^+$-ATPase). Distances are edlib infix edit distances including any intron, so they are comparable with each other but only approximately with those thresholds. Records are joined to species through the reference culture names; records whose GenBank name disagrees with the reference name of their culture are left out (6).

Table S7: Single-linkage clusters of the sequences of the 37 named species, within each order, at two identity thresholds: all retained sequences, and main-type sequences only. Short only and same length only: clusters made only of short-5.8S or only of same-length divergent sequences; several species: clusters holding sequences of more than one species. Exploratory; no rule depends on it.

| | | Clusters | | Clusters of all sequences | | |
|---|---|---|---|---|---|---|
| Region | Identity | all sequences | main type | short only | same length only | several species |
| Full ITS | 98.5% | 240 | 220 | 18 | 2 | 2 |
| Full ITS | 97% | 110 | 96 | 12 | 2 | 2 |
| Krüger fragment | 98.5% | 110 | 95 | 13 | 2 | 1 |
| Krüger fragment | 97% | 56 | 46 | 8 | 2 | 4 |
| Whole sequence | 98.5% | 71 | 58 | 11 | 2 | 3 |
| Whole sequence | 97% | 36 | 30 | 5 | 1 | 7 |

### Supplementary Data S1

Alignments and maximum-likelihood trees of the culture sequences, for each order and for the SSU region, full ITS, LSU region and concatenated sequence: alignments (FASTA), trees (Newick, with SH-aLRT and ultrafast bootstrap supports on the branches, and tips labelled with accession, species, culture, sequence class and number of identical sequences), the sequences each tip stands for, and the partitions of the concatenated alignment (`benchmarks/copy_checks/supplementary_data_S1`).

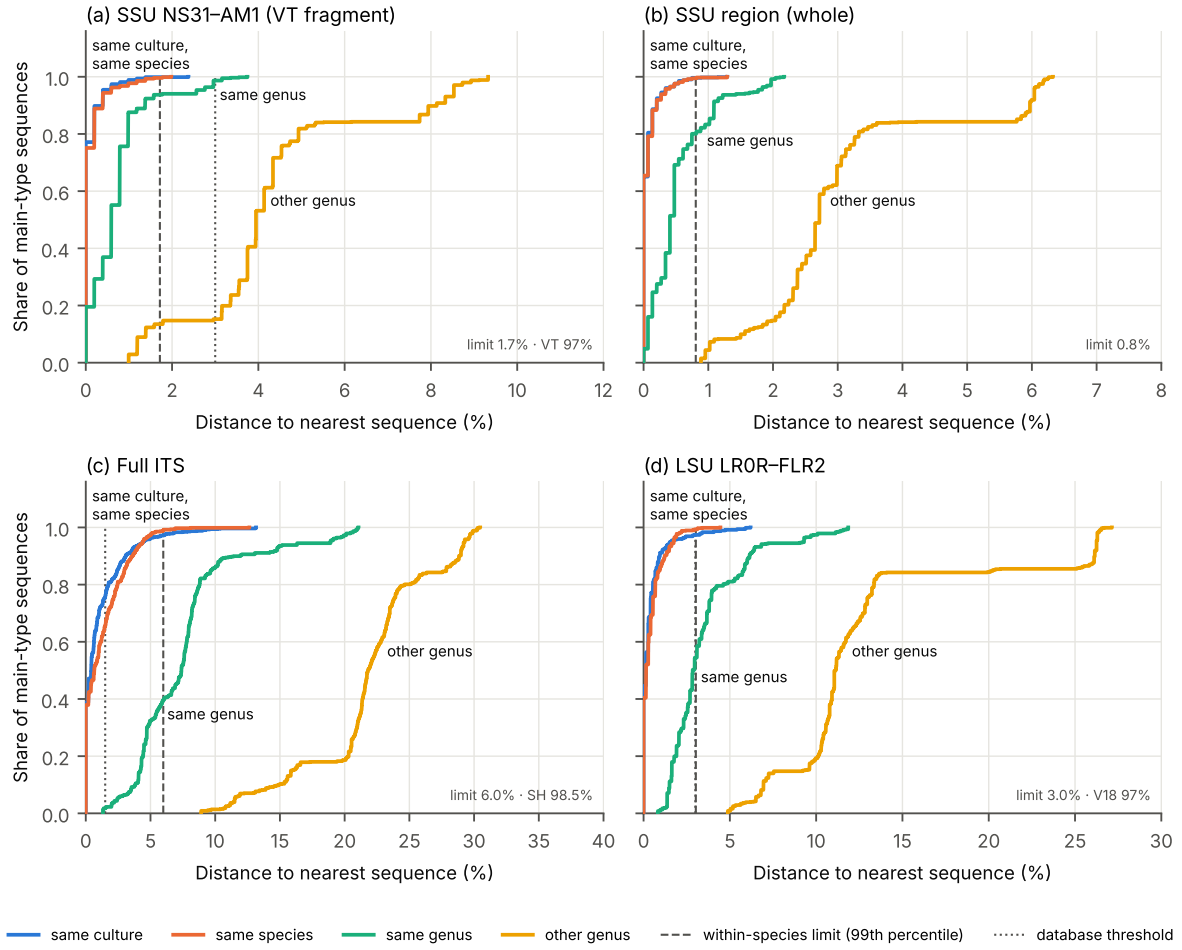



Figure S1: Distances between closely related sequences at four levels of relatedness. For each main-type sequence, the distance to its nearest main-type sequence in the same culture, in another culture of the same species, in another species of the same genus, and in another genus of the same order. (a) The NS31–AM1 fragment, on which MaarjAM VTs are defined; (b) the whole SSU region; (c) the full ITS; (d) the LSU fragment LR0R–FLR2. Dashed lines, the within-species limit (the 99th percentile of the distance between cultures of one species); dotted lines, the identity thresholds of the databases on the fragments to which they apply (VT 97%, SH 98.5%, V18 97%). The Paraglomerales sequences lack the NS31–AM1 fragment.

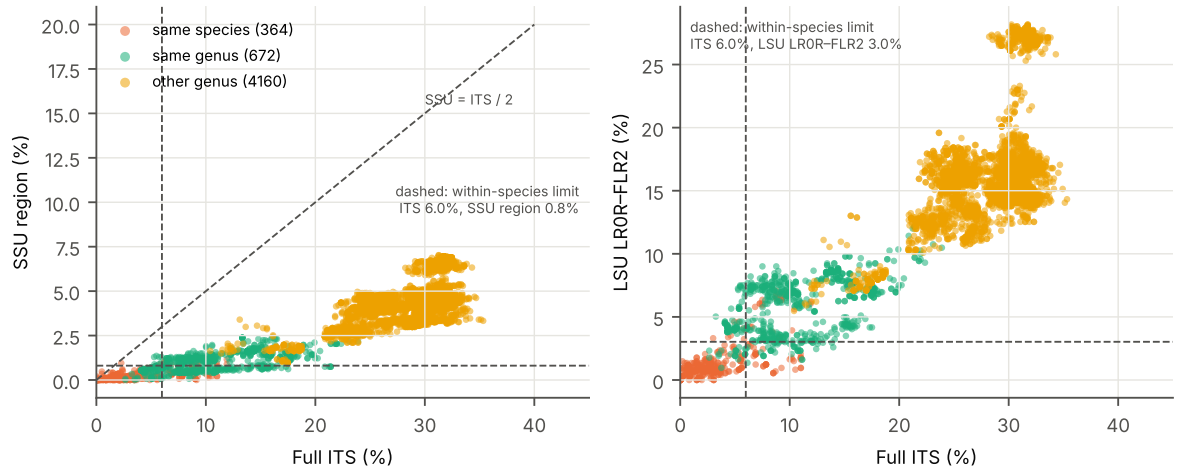



Figure S2: SSU and LSU distances compared with ITS distances between pairs of cultures. Distances between the medoid sequences of pairs of cultures within an order: (a) SSU region and (b) LSU fragment LR0R–FLR2, against the full ITS, coloured by relatedness (numbers of pairs in the legend). Dashed lines, the within-species limits; in (a), the line SSU = ITS/2, above which the SSU-discordance test applies once the SSU differs by at least 1%.

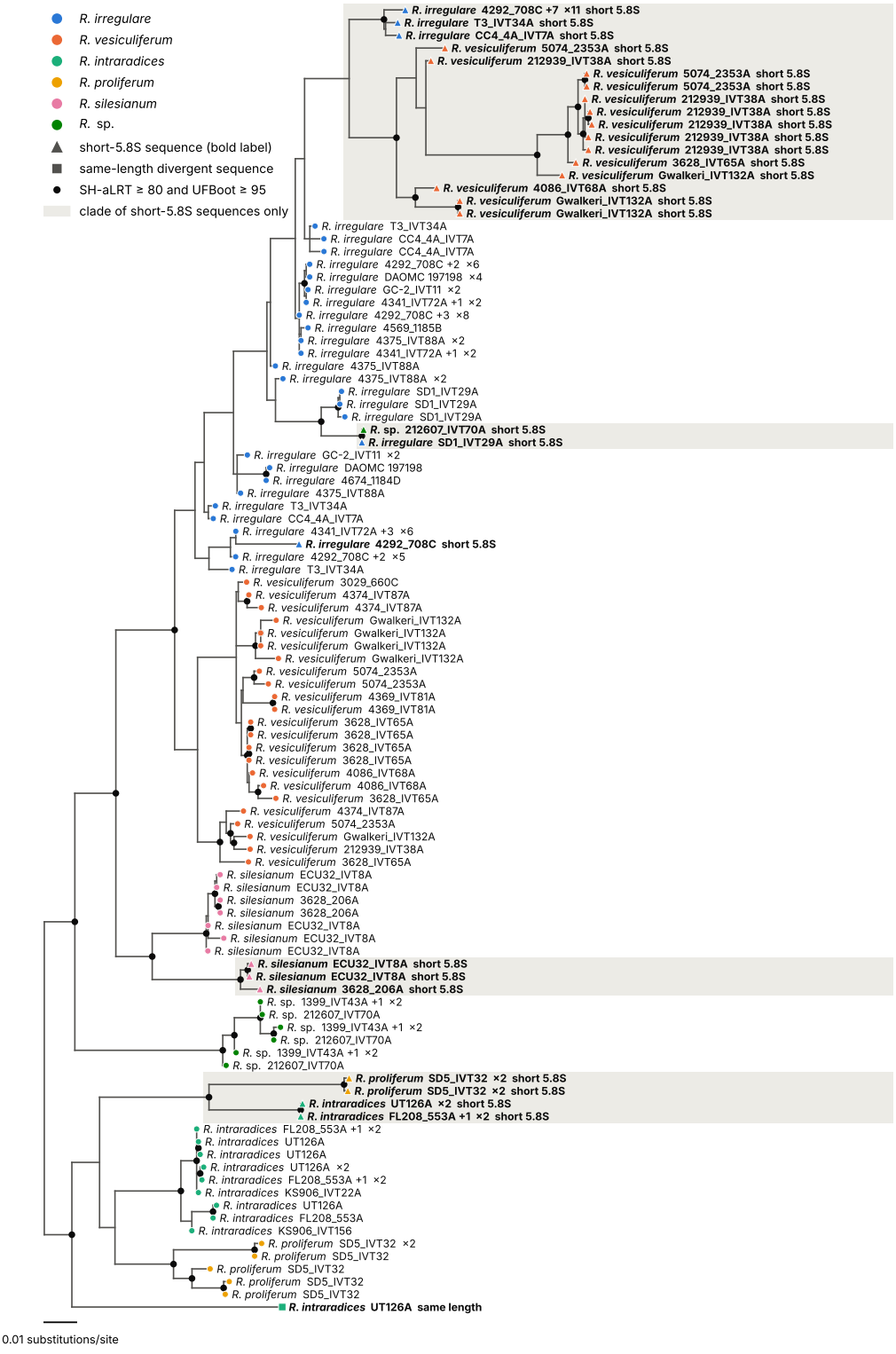



Figure S3: Uncollapsed phylogeny of the *Rhizoglomus* clade of Figure 3. Tips are distinct sequences, labelled with the species (*R.*, *Rhizoglomus*) and the culture; '$+n$', the same sequence also occurs in $n$ further cultures; '$\times n$', the number of identical sequences. Triangles and bold labels, short-5.8S sequences; square, the same-length sequence; black dots, branches with SH-aLRT $\geq 80$ and ultrafast bootstrap $\geq 95$; shading, clades made only of short-5.8S sequences.